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Return insurance in a competitive market: Benefiting the high-quality or low-quality retailer?

Hui Xiong, Jie Liu *

School of Management, Huazhong University of Science and Technology, Wuhan, Hubei 430074, People's Republic of China

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ABSTRACT

Return shipping insurance (return insurance), as emerging insurance on many online platforms (e.g., Taobao.com and JD.com), develops rapidly in China over the past decade. Many retailers on the platforms purchase return insurance for consumers to reduce consumers' hassle cost, while some retailers abandon return insurance. To investigate the retailers' divergence on the adoption of the return insurance, we build a duopoly model in which two competing online retailers with different quality levels sell products to consumers who have uncertainty about their satisfaction. Both retailers can choose whether to purchase return insurance at different premiums. Our finding shows that as long as the increase in the consumers' expected utility is higher than the cost of premium, the retailer is better off offering return insurance, independent of his competitor's decision. Then, when the premium for the high-quality product is sufficiently low and the premium for the low-quality product is sufficiently high, the retailer with the high-quality product purchases return insurance for consumers and the retailer with the low-quality product abandons return insurance. Moreover, when both retailers offer return insurance, the retailer with the high-quality product may reduce or increase the retail price and the retailer with the low-quality product always increases the retail price. Finally, we examine the interaction between the retailers' decisions on quality levels and the adoption of return insurance, and show that both retailers may improve quality levels when offering return insurance.

1. Introduction

Return shipping insurance (return insurance for short), as emerging insurance on many online platforms (e.g., Taobao.com and JD.com), develops rapidly in China over the past decade. The total sales of the return insurance premiums in China reach RMB 8.27 billion in the first half of 2022.[1] Recently, return insurance has been adopted by an online retailer who sells laptop screens in the United States (Zhang et al., 2022).[2] The retailers on the platforms offer return insurance for consumers to alleviate their concerns about the necessity to pay the return shipping fee (Li et al., 2021). When a return occurs, the consumer can obtain a compensation for the shipping fee from the insurance provider.

For the retailers, purchasing return insurance for consumers enables them to attract more consumers and increase the retail prices (e.g., Fan and Chen, 2020; Li et al., 2018, 2021; Zhang et al., 2022). However, if a retailer offers return insurance, the retailer should pay a premium for each product sold, which would be a significant financial expenditure. In addition, the return insurance may lead to more product returns as the demand increases, which exhausts the retailer (Fan and Chen, 2020; Geng and Li, 2017). Consequently, retailers make different choices, regarding return insurance. For example, both Apple and Oppo are cellphone retailers on Tmall.com. Apple, as a high-quality retailer, does not offer return insurance for consumers; while Oppo with lower quality offers return insurance. The quality levels are inferred from the retailers' ratings, following Zhang et al. (2022).[3] Besides, among the retailers who sell TVs on the e-commerce platform Pinduoduo in China, Xiaomi and KONKA purchase return insurance for consumers, while TCL and Hisense do not offer return insurance.[4]

In the aforementioned examples, the retailers in an identical product category make different choices on purchasing return insurance for consumers. Existing work shows that return insurance can benefit a

* Corresponding author.
E-mail addresses: hxionghust@163.com (H. Xiong), victoriajlu@163.com (J. Liu).

[1] Please refer to the report called "Internet Property Insurance Development Analysis Report in the first half of 2022" through the following link: http://www.iachina.cn/art/2022/9/27/art_22_106400.html.

[2] Please refer to the following link: https://www.laptopscreen.com/blog/return-shipping-insurance-and-why-do-i-need-it/.

[3] Regarding the retailers' ratings and return insurance strategies, please refer to the following links: https://oppo.tmall.com/ and https://apple.tmall.com/.

[4] Since the platform, Pinduoduo, only allows consumers to purchase products through the PDD app, we provide relevant screenshots in Appendix C.

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monopoly retailer (e.g., Chen et al., 2021; Fan and Chen, 2020; Li et al., 2021), while the impacts of return insurance on retailers and consumers in a competitive market are not clear. In this paper, we aim to provide a model to investigate the retailers' decisions on return insurance in a competing environment. Our study intends to answer the following questions: (1) When should the competing retailers purchase return insurance for consumers? (2) How does the return insurance strategy impact the retailers' prices, demand, and profit? (3) How do the retailers' quality decisions interact with the adoption of return insurance?

To address these research questions, we build a duopoly model in which two competing online retailers sell products with different quality levels to consumers. Every consumer (she) has uncertainty about her satisfaction with the products. Following the literature (e.g., Chen and Chen, 2017; McWilliams, 2012), consumer satisfaction rate is influenced by the quality level, that is, the high (low) quality product has a high (low) satisfaction rate. Moreover, the return rate for the product with a high satisfaction rate is lower, compared to the product with a low satisfaction rate. Facing these consumers, two retailers offer different quality levels. The retailer offering a high (low) quality product is called the high-quality (low-quality) retailer. The quality decision affects consumers' satisfaction rates. Moreover, each retailer (he) decides whether to pay a premium to purchase return insurance for consumers. In line with reality, the retailer with a high return rate (i.e., the low-quality retailer) has to pay a higher return insurance premium, compared to the high-quality retailer.[5] The degree of the return insurance premium difference between retailers is labeled "the premium differentiation degree".

We find that as long as the increase in the consumers' expected utility is higher than the cost of premium, the retailer is better off offering return insurance, independent of his competitor's decision. Then, only the high-quality retailer purchases return insurance for consumers when the premium for the high-quality retailer is sufficiently low and the premium for the low-quality retailer is sufficiently high. In this case, the return insurance enables the high-quality retailer to obtain more consumers. However, for the low-quality retailer, the premium is costly and the additional profit brought by the insurance cannot counterbalance the cost of premiums. Thus, the low-quality retailer gives up buying return insurance. Our finding contrasts with the results in Zhang et al. (2022), which does not identify this equilibrium in the full-information case.

Moreover, we show that when the premiums for the high-quality retailer and the low-quality retailer are low, both retailers offer return insurance. Then, the low-quality retailer increases the retail price, while the high-quality retailer may increase or reduce the retail price. When the premium for the low-quality retailer is relatively low, the low-quality retailer can increase the retail price by a relatively small amount to serve the consumers with moderate valuation. Then, the high-quality retailer has an incentive to reduce the difference between his retail price and his competitor's retail price to retain some consumers; otherwise, the consumers may switch from the high-quality product to the low-quality product. Our result contrasts with the finding in the literature. Fan and Chen (2020), Li et al. (2018, 2021) and Zhang et al. (2022) demonstrate that the adoption of return insurance leads the retailer to increase the retail price.

Furthermore, we find that when both retailers offer return insurance, they have more demand if the premium differentiation degree is moderate. In this case, although both retailers increase the retail prices, return insurance enables the consumers to obtain more utility and more consumers purchase the products. Thus, the demand for two retailers increases, and the adoption of return insurance leads to the Pareto improvement. However, if the premium differentiation degree is relatively large, the low-quality retailer further increases his retail price to cover the expense of offering return insurance. Then, more consumers switch from the low-quality product to the high-quality product and the low-quality retailer faces a reduction in demand.

Finally, we extend our basic model to the following two scenarios: (1) online retailers endogenously determine the quality levels; (2) consumers' valuation is influenced by the quality. First, in the scenario with endogenous quality decisions, we observe that when the return insurance provides enough value (i.e., the shipping compensation is high or the premium is low), both retailers choose to purchase return insurance for consumers. Moreover, when the premium function is convex, both retailers may improve their quality levels to increase satisfaction rates. Fan and Chen (2020), Li et al. (2018), and Zhang et al. (2022) show that when consumers are aware of the product quality, the low-quality retailer with a low satisfaction rate can benefit more from the return insurance. However, in our model, the convex premium function enables a retailer with a higher quality level (satisfaction rate) to take more advantage of return insurance. Our finding suggests that the insurance providers can adjust the insurance premium scheme to encourage the retailers to increase their quality levels. Second, we consider the case where the consumers' valuation is influenced by the quality. In this case, the retailers are more likely to benefit from offering return insurance. Compared to the basic model, the range in which the adoption of return insurance enables two retailers to increase their demand is larger. Consumer heterogeneity increases the efficiency realized from purchasing return insurance. Thus, both retailers can have more demand and benefit from offering return insurance.

Our contribution includes the following three aspects. (1) We consider a scenario in which competing retailers can choose to buy return insurance and consumers cannot purchase insurance by themselves, which is absent in the existing papers (e.g., Chen et al., 2022; Li et al., 2018). In reality, the e-commerce platform Pinduoduo in China only allows retailers to purchase return insurance for consumers and does not allow consumers to buy.[6] (2) To the best of our knowledge, we are the first to consider the interaction between the retailers' decisions on quality levels and the offering of return insurance. The quality decision affects the satisfaction rate and the product acquisition cost, which attracts much attention in marketing and operation management areas (e.g., Chen and Chen, 2017; Huang et al., 2018). In our study, the quality decision also influences the costs in paying the premiums. (3) Our study provides several managerial implications. For example, we identify the conditions under which the high-quality retailer should reduce the retail price. Moreover, we suggest that the insurance providers can adjust the insurance premium scheme to encourage the retailers to increase their quality levels.

The remainder of the paper is organized as follows. Section 2 provides a summary of the literature relevant to our work. Section 3 presents the model setup. Section 4 then lays out the analysis. In Section 5, we extend our basic model to the following two scenarios: (1) online retailers endogenously determine the quality levels; (2) consumers' valuation is influenced by the quality. Finally, we conclude in Section 6. All proofs are relegated to the appendix.

2. Literature review

Our paper is related to the literature on product returns. Many studies (e.g., Chu et al., 1998; Nageswaran et al., 2020; Su, 2009) discuss the optimal refund for the return policy (i.e., full refund, partial refund, or no refund). For example, Nageswaran et al. (2020) study an omnichannel retailer's optimal decisions on the return policy and price. Su (2009) investigates the influence of return policies on supply chain

[5] Please refer to the related data about return insurance premiums on Taobao and Tmall through the following link: https://www.ebrun.com/ebrungo/zb/434849.shtml.

[6] Please refer to the following link: https://www.chinatesting.com.cn/caijing/202206/68632.html.

performance. Shulman et al. (2009) examine the return policies and combine them with the information provision strategies. Among the literature on return policy, many papers focus on a full-refund return policy, such as money back guarantee (MBG). For example, Moorthy and Srinivasan (1995) show that MGB can signal high quality to consumers and a retailer with high quality can benefit from adopting an MBG, which focus on the return policy in a monopolistic environment. There are some studies examining MBG in a competitive market. McWilliams (2012) finds that when both retailers offer MBGs to consumers who are fully informed about the product quality levels, the low-quality retailer gains more benefits, compared to the high-quality retailer. In contrast with McWilliams (2012) and Huang et al. (2018) consider asymmetric salvage values of products and prove that the low-quality retailer is hurt by the MBG while the high-quality retailer benefits from the MBG. Following the studies of McWilliams (2012) and Huang et al. (2018), our paper explores a competitive environment with a high-quality retailer and a low-quality retailer. Consumers know the product quality before making a purchase. Differently, we focus on the retailers' decisions on the offering of return insurance.

Our paper belongs to a line of return insurance literature. Fan and Chen (2020) show that the online retailer purchases return insurance for consumers when the premium is low and the compensation is high. Chen et al. (2021) examine the relationship between the decisions on the return insurance offering and the selling formats (i.e., agency selling or reselling). They find that whether to purchase return insurance is influenced by the return loss and premium. Li et al. (2021) consider two types of return insurance strategies, in which the retailer purchases insurance for consumers or consumers buy the insurance for themselves. They find that the return insurance enables the retailer to gain more profit and the social welfare is increased. Geng and Li (2017) and Zhang et al. (2022) investigate the informational role of return insurance and propose that the return insurance can convey a signal of high quality to consumers who are unaware of the product quality. Geng et al. (2017) study the insurer's optimal pricing strategy on the insurance premium and compensation. These studies investigate the adoption of return insurance when the retailer is monopolistic. In contrast, Li et al. (2018) assume that two competing retailers provide return insurance for consumers, but do not explore their purchasing decisions on return insurance.

Our work is closer to the study by Chen et al. (2022), who consider two competing retailers' choices on offering return insurance when consumers may purchase the return insurance. Compared to Chen et al. (2022), consumers cannot purchase insurance in our model, which is in line with reality. For example, the e-commerce platform Pinduoduo in China only allows retailers to purchase return insurance and does not allow consumers to buy the insurance. Moreover, we consider the retailers' endogenous quality decisions and examine the interaction between their decisions on the quality and the offering of return insurance. We show that under a convex premium function, both retailers may purchase return insurance for consumers and improve their quality levels.

3. Model

Competing retailers. We consider a scenario in which a high-quality retailer (retailer H) sells a product with high quality q_H (product H) and a low-quality retailer (retailer L) sells a product with low quality q_L (product L) to consumers on an online platform, where $q_H > q_L$. For retailer/product i ($i \in \{L, H\}$), p_i represents the retail price, C_i denotes the unit product acquisition cost, S_i ($S_i < C_i$) represents the salvage cost of a returned product, and T_i denotes the transaction cost of the retailer for a returned product. Moreover, we assume that the quality levels are exogenous in Sections 4 and 5.2, following the studies by McWilliams (2012) and Huang et al. (2018). In Section 5.1, we consider the scenario with endogenous quality levels. We summarize the notations in Table 1.

Table 1
Summary of frequently used notations.

	Explanation
Sections 3–5	
q_i	Quality level for product i, $i = \{L, H\}$
γ_i	Satisfaction rate for product i and $\gamma_i = g(q_i)$, $i = \{L, H\}$
p_i, D_i, C_i	Retail price, demand, and unit product acquisition cost for product i, $i = \{L, H\}$
S_i, T_i	Salvage value and transaction cost of the retailer for a returned product i, $i = \{L, H\}$
k_i	Retailer i's decision on whether to offer the return insurance, $i = \{L, H\}$; $k_i = 0$ or 1 corresponds to non-adoption or adoption of return insurance
K	Return insurance strategy combination for retailers L and H, $K = k_L k_H$
K^*	Equilibrium insurance strategy combination, $K^* \in \{00, 01, 10, 11\}$
f_i	Return insurance premium for product i, $i = \{L, H\}$
β	Premium differentiation degree, and $\beta = \frac{f_L - f_H}{\gamma_H - \gamma_L} > 0$
r	Consumers' hassle cost of returning a product
v	Consumers' product evaluation, $v \sim U[0, 1]$
w	Compensation obtained by the consumer if the return occurs
Section 5.1	
α	Product acquisition coefficient
ξ	Premium coefficient
Section 5.2	
θ	Consumers' willingness to pay forquality, $\theta \sim U[0, 1]$

Consumer satisfaction and returns. Following the literature on consumer returns (e.g., Chen and Chen, 2017; McWilliams, 2012; Zhang et al., 2022), consumer satisfaction rate is influenced by the quality level, and high quality reflects a high satisfaction rate. We assume that retailer i ($i \in \{L, H\}$) has a consumer satisfaction rate $g(q_i)$ and $g'(\cdot) > 0$. The satisfaction rate $g(q_i)$ is the probability that consumers are satisfied with product i and keep it. Conversely, $1 - g(q_i)$ is the consumer return rate for retailer i. For simplicity, we let γ_i denote $g(q_i)$ in the scenarios with exogenous quality levels in Sections 4 and 5.2, and $\gamma_H > \gamma_L$ since $q_H > q_L$.

Moreover, if a consumer is satisfied with product i, the consumer's valuation is v, where v is uniformly distributed between 0 and 1 (i.e., $v \sim U[0, 1]$). In Section 5.2, we consider the scenario in which the valuation is influenced by the quality. If the consumer is not satisfied with the product i, the consumer's valuation is zero.

Hassle cost and return insurance. Consumers incur a hassle cost r, which includes time, effort, and delivery fee, to return a product. Moreover, when the retailer provides the return insurance for consumers, each consumer can obtain a compensation w ($w < r$) for each return. Then, $r - w > 0$ represents the hassle cost that is not covered by the compensation.[7] Note that following the relevant studies on consumer returns (e.g., Chen and Chen, 2017; McWilliams, 2012; Zhang et al., 2022), we do not consider the forward shipping fee.[8]

Each retailer can choose whether to purchase return insurance at a premium f_i for each consumer. The insurance can alleviate the consumer's hassle cost if the consumer is not satisfied with the product and wants to return it. Let $k_i \in \{0, 1\}$ denote the retailer i's decision on whether to offer the return insurance, in which $k_i = 0$ or 1 represents non-adoption or adoption of return insurance. Then, $\{k_L, k_H\}$ is the return insurance strategy combination of retailer L and retailer H. When $k_L = k_H = 0$, $\{k_L, k_H\} = \{0, 0\}$. We use $K = 00$ to represent $\{k_L, k_H\} = \{0, 0\}$. Similarly, we can write K for the other three possible return insurance strategy combinations, and let K^* denote the equilibrium result. Thus, $K \in \{00, 01, 10, 11\}$ and $K^* \in \{00, 01, 10, 11\}$.

[7] In general, the compensation is determined by the consumer and the retailer's locations. To simplify the analysis, we assume that the compensations are identical for both retailers, following the literature (e.g., Chen et al., 2022; Zhang et al., 2022).

[8] If we consider the forward shipping fee, the equilibrium results remain qualitatively similar to those of our paper. We can provide the detailed analysis upon request.

Moreover, we assume that $f_H < f_L < w$, which is in line with the reality that the retailer should pay a higher return insurance premium for a higher consumer return rate and the premium is lower than the compensation (e.g., Fan and Chen, 2020; Geng and Li, 2017; Geng et al., 2017). Let $\beta = \frac{f_L - f_H}{\gamma_H - \gamma_L} > 0$ denote the degree of the return insurance premium difference between the low-quality retailer and high-quality retailer and we call it "premium differentiation degree". The introduction of β allows us to simplify $f_H < f_L$ with $\beta > 0$, and enables us to characterize the equilibrium results by capturing the exact difference in return insurance premiums between retailers L and H.

Consumers' utility: Each consumer buys at most one product. Then, we obtain that the consumers' expected utility of purchasing product i is

$$EU_i^K = \gamma_i(v - p_i^K) - (1 - \gamma_i)(r - k_i w). \tag{1}$$

Retailers' demand and profit: A consumer chooses to buy product i when the expected utility EU_i^K is non-negative. Besides, a consumer chooses to buy product L rather than product H when the expected utility $EU_L^K > EU_H^K$, and vice versa. Then, if $EU_L^K > EU_H^K$ and $EU_L^K \geq 0$, the consumer will buy product L; else if $EU_L^K < EU_H^K$ and $EU_H^K \geq 0$, the consumer will buy product H; otherwise, the consumer will give up buying either product. Given $v \sim U[0,1]$, we can derive the demand for retailers L and H as follows:

$$(D_L^K,\ D_H^K) = \begin{cases} (0, 1 - v_H^K) & if\ p_H^K \leq \underline{p}_H^K; \\ (v_{HL}^K - v_L^K, 1 - v_{HL}^K) & if\ \underline{p}_H^K < p_H^K < \overline{p}_H^K; \\ (1 - v_L^K, 0) & if\ p_H^K \geq \overline{p}_H^K, \end{cases} \tag{2}$$

where v_H^K is computed from $EU_H^K(v) = 0$, v_{HL}^K is obtained from $EU_L^K(v) = EU_H^K(v)$, and v_L^K is derived from $EU_L^K(v) = 0$. In addition, from $v_{HL}^K = 1$ and $v_{HL}^K = v_L^K$, we can obtain $\overline{p}_H^K$ and $\underline{p}_H^K$, between which (i.e., $\underline{p}_H^K < p_H^K < \overline{p}_H^K$) both retailers have positive sales. It also means that retailer L is driven out of the market if $p_H \leq \underline{p}_H^K$, and retailer H is driven out of the market if $p_H \geq \overline{p}_H^K$. To investigate two competing retailers' return insurance strategies, we focus on the case where both retailers have positive sales (co-exist). Therefore, given the condition that $\underline{p}_H^K < p_H < \overline{p}_H^K$, $D_L^K = v_{HL}^K - v_L^K$, $D_H^K = 1 - v_{HL}^K$.

Moreover, the profit for retailer i is

$$\pi_i^K = D_i^K[\gamma_i p_i^K + (1 - \gamma_i)(S_i - T_i) - C_i - k_i f_i]. \tag{3}$$

Sequence of events: The sequence of events is summarized as follows. (1) Each retailer decides whether to offer return insurance, and sets the retail price p_i. (2) Consumers make purchasing decisions. (3) If a consumer is dissatisfied with the product, she returns the product.

4. Analysis

In this section, we characterize the retailers' pricing strategies in four cases distinguished by their decisions on return insurance. Then, we compare the retailers' profit and obtain their optimal return insurance strategies.

4.1. Pricing strategies

At stage 2, consumers make purchasing decisions, and two retailers' demand is shown in Eq. (2). At stage 1, both online retailers decide the retail prices and their insurance strategies, and their optimization problems are written as follows.

$$\max_{p_i^K} \pi_i^K = D_i^K[\gamma_i p_i^K + (1 - \gamma_i)(S_i - T_i) - C_i - k_i f_i]. \tag{4}$$

Then, we can solve the optimization problems and characterize the retailers' optimal pricing strategies. For the sake of brevity, we relegate the analysis to Appendix A. Following the study by Chen and Chen (2017), we define $\Delta_i^K = \gamma_i - C_i + (1 - \gamma_i)(S_i - T_i - r + k_i w) - k_i f_i$ as the *maximum net shared value* of retailer i, where $i = \{L, H\}$. Note that Δ_i^K refers to the consumer's highest net valuation for product i ($v = 1$ in this case). Next, we present retailers' pricing strategies, demand, and profit in Lemma 1. The values of Δ_i^K are summarized in Table 2.

Lemma 1. *When* $\frac{\gamma_H}{2\gamma_H - \gamma_L} < \frac{\Delta_H^K}{\Delta_L^K} < \frac{2\gamma_H - \gamma_L}{\gamma_L}$, *retailers' optimal retail prices in case K are*

$$\begin{cases} p_L^K = 1 - \frac{\gamma_L \Delta_H^K + 2\gamma_H \Delta_L^K}{\gamma_L(4\gamma_H - \gamma_L)} - \frac{(1-\gamma_L)(r - k_L w)}{\gamma_L}, \\ p_H^K = 1 - \frac{2\Delta_H^K + \Delta_L^K}{4\gamma_H - \gamma_L} - \frac{(1-\gamma_H)(r - k_H w)}{\gamma_H}. \end{cases} \tag{5}$$

Moreover, their demand and profit are shown as follows:

$$D_L^K = \frac{\gamma_H[(2\gamma_H - \gamma_L)\Delta_L^K - \gamma_L \Delta_H^K]}{\gamma_L(4\gamma_H - \gamma_L)(\gamma_H - \gamma_L)},\ \text{and}\ D_H^K = \frac{(2\gamma_H - \gamma_L)\Delta_H^K - \gamma_H \Delta_L^K}{(4\gamma_H - \gamma_L)(\gamma_H - \gamma_L)}, \tag{6}$$

$$\pi_L^K = \frac{\gamma_H[(2\gamma_H - \gamma_L)\Delta_L^K - \gamma_L \Delta_H^K]^2}{\gamma_L(4\gamma_H - \gamma_L)^2(\gamma_H - \gamma_L)},\ \text{and}\ \pi_H^K = \frac{[(2\gamma_H - \gamma_L)\Delta_H^K - \gamma_H \Delta_L^K]^2}{(4\gamma_H - \gamma_L)^2(\gamma_H - \gamma_L)}, \tag{7}$$

where $K \in \{00, 01, 10, 11\}$.

Lemma 1 characterizes the optimal price p_i^K, the corresponding demand D_i^K, and the profit π_i^K. It is worth mentioning that two competing retailers co-exist under the condition that $\frac{\gamma_H}{2\gamma_H - \gamma_L} < \frac{\Delta_H^K}{\Delta_L^K} < \frac{2\gamma_H - \gamma_L}{\gamma_L}$; otherwise, one retailer is driven out of the market. Specifically, if $\frac{\Delta_H^K}{\Delta_L^K} \leq \frac{\gamma_H}{2\gamma_H - \gamma_L}$ (or $\frac{\Delta_H^K}{\Delta_L^K} \geq \frac{2\gamma_H - \gamma_L}{\gamma_L}$), retailer H (or retailer L) has no sales. Then, we compare the profit in different cases and summarize the results in the next section.

4.2. Return insurance strategies

In this section, we characterize the retailers' optimal return insurance strategies. Specifically, as in Lemma 1, we present the retailers' profit in four cases. Then, we compare the profit and describe the retailers' decisions on the return insurance strategy in the following propositions. All the proofs are relegated to Appendix A.

Proposition 1. *Retailer* i ($i \in \{L, H\}$) *purchases return insurance for consumers (i.e.,* $k_i = 1$*) if and only if* $(1 - \gamma_i)w - f_i \geq 0$,[9] *independent of his competitor's decision. Then, the following combinations of retailers' return insurance strategies can emerge as market equilibria:*

1. *When* $f_H \leq (1 - \gamma_H)w$,
   a. *if* $f_L \leq (1 - \gamma_L)w$, *both retailer L and retailer H purchase return insurance (i.e.,* $K^* = 11$*);*
   b. *if* $f_L > (1 - \gamma_L)w$, *retailer L does not purchase return insurance, while retailer H does (i.e.,* $K^* = 01$*).*
2. *When* $f_H > (1 - \gamma_H)w$,
   a. *if* $f_L \leq (1 - \gamma_L)w$, *retailer L purchases return insurance, while retailer H does not (i.e.,* $K^* = 10$*);*
   b. *if* $f_L > (1 - \gamma_L)w$, *retailer L and retailer H never purchase return insurance (i.e.,* $K^* = 00$*).*

[9] Note that the profitability of the insurance policy for the insurer depends on condition $(1 - \gamma_i)w - f_i < 0$ but that the insurer may accept negative profit from selling insurance to expand the market and improve the profit from the investments of the premiums. As reported, the total premiums for return insurance are large, and the premium income reached RMB 8.27 billion in the first half of 2022 (Please refer to the link: http://www.iachina.cn/art/2022/9/27/art_22_106400.html). The huge amounts of return insurance premiums may generate substantial investment profit for the insurer.

Table 2
The values of Δ_i^K, $i = \{L, H\}$ and $K = \{00, 01, 10, 11\}$.

	Δ_L^K	Δ_H^K
$K = 00$	$\gamma_L - C_L + (1-\gamma_L)(S_L - T_L - r)$	$\gamma_H - C_H + (1-\gamma_H)(S_H - T_H - r)$
$K = 01$	$\gamma_L - C_L + (1-\gamma_L)(S_L - T_L - r)$	$\gamma_H - C_H + (1-\gamma_H)(S_H - T_H - r + w) - f_H$
$K = 10$	$\gamma_L - C_L + (1-\gamma_L)(S_L - T_L - r + w) - f_L$	$\gamma_H - C_H + (1-\gamma_H)(S_H - T_H - r)$
$K = 11$	$\gamma_L - C_L + (1-\gamma_L)(S_L - T_L - r + w) - f_L$	$\gamma_H - C_H + (1-\gamma_H)(S_H - T_H - r + w) - f_H$

Proposition 1 demonstrates that as long as the increase in the consumers' expected utility is higher than the cost of premium (i.e., $(1-\gamma_i)w - f_i \geq 0$), retailer i ($i \in \{L, H\}$) is better off offering return insurance, independent of his competitor's decision. Intuitively, a low premium f_i induces retailer i to purchase return insurance since the insurance reduces the consumers' hassle cost and more consumers are willing to make a purchase. Then, when the premiums for retailer H and L are low (i.e., $f_i \leq (1-\gamma_i)w$, $i \in \{L, H\}$), both retailers H and L purchase return insurance. On the contrary, when the premiums for retailer H and L are high (i.e., $f_i > (1-\gamma_i)w$, $i \in \{L, H\}$), no retailer purchases return insurance.

Moreover, only retailer H purchases return insurance when the premium for retailer H is sufficiently low (i.e., $f_H \leq (1-\gamma_H)w$) and the premium for retailer L is sufficiently high (i.e., $f_L > (1-\gamma_L)w$). In this case, the return insurance enables retailer H to obtain more consumers and the low premium results in a low cost of offering return insurance. However, for retailer L, the premium is costly and the additional profit brought by the insurance cannot counterbalance the cost of premiums. Thus, retailer L gives up buying return insurance. Our finding contrasts with the results of the full-information case in the study by Zhang et al. (2022). Zhang et al. (2022) consider a single retailer offering either a high-quality or low-quality product and assume that the premiums are identical for both quality levels. They show that only the low-quality retailer purchases insurance when the premium is moderate. In contrast, we consider two competing retailers' decisions on offering return insurance and assume that the premium for the low-quality retailer is higher than that for the high-quality retailer, which is in line with the practice. In our model, only retailer L purchases return insurance when the premiums for both retailers are high (i.e., $f_L > f_H > (1-\gamma_H)w$) and the premium for retailer L is lower than or equal to the threshold (i.e., $f_L \leq (1-\gamma_L)w$).

Next, to investigate the impacts of two competing retailers' choices of the return insurance adoption, we introduce the premium differentiation degree β to capture the exact difference in return insurance premiums between retailers L and H, and examine the impact of premium differentiation degree β on the equilibrium results. Since $\beta = \frac{f_L - f_H}{\gamma_H - \gamma_L}$, we replace f_L with $\beta(\gamma_H - \gamma_L) + f_H$ in the following discussion. As a result, a larger positive β reflects the fact that retailer L should afford a higher premium than retailer H, given f_H and $(\gamma_H - \gamma_L)$.

Proposition 2. *Regarding the retail prices, we have the following results:*

1. When only one retailer offers return insurance (i.e., $K^* = 01$ or 10), the retailer with return insurance charges a higher retail price than the price in case $K = 00$ (i.e., $p_L^{10} > p_L^{00}$ and $p_H^{01} > p_H^{00}$). However, his competitor has to offer a lower price (i.e., $p_H^{10} \leq p_H^{00}$ and $p_L^{01} \leq p_L^{00}$).

2. When both retailers offer return insurance (i.e., $K^* = 11$), retailer L offers a higher retail price than the price in case $K = 00$ (i.e., $p_L^{11} > p_L^{00}$), and

a. if $\beta \leq (\frac{(2\gamma_H - 1)w}{\gamma_H} - \frac{3f_H}{\gamma_H - \gamma_L})^+$, retailer H charges a lower retail price than the price in case $K = 00$ (i.e., $p_H^{11} \leq p_H^{00}$);

b. if $(\frac{(2\gamma_H - 1)w}{\gamma_H} - \frac{3f_H}{\gamma_H - \gamma_L})^+ < \beta < \frac{(1-\gamma_L)w - f_H}{\gamma_H - \gamma_L}$, retailer H offers a higher retail price than the price in case $K = 00$ (i.e., $p_H^{11} > p_H^{00}$).

Proposition 2 shows that when only one retailer offers return insurance, the retailer with return insurance increases the retail price. This finding is in line with the previous studies (e.g., Fan and Chen, 2020; Li et al., 2021; Zhang et al., 2022), which demonstrate that the retailer increases the retail price to cover additional marginal costs caused by premiums and improve his profit when offering return insurance. In contrast, we consider the retail competition and find that a retailer with return insurance may reduce the retail price. Specifically, when the premium for retailer H is low and the premium differentiation degree is sufficiently small (i.e., $\beta \leq (\frac{(2\gamma_H - 1)w}{\gamma_H} - \frac{3f_H}{\gamma_H - \gamma_L})^+$), both retailers offer return insurance and retailer H reduces the retail price. The additional expected utilities from offering return insurance for product H are smaller than those for product L, since the negative impact of receiving a dissatisfied product is reduced more for product L. Then, retailer L is easier to obtain more consumers by offering return insurance. When the premium for retailer H is low and the premium differentiation degree is sufficiently small, the premium for retailer L is low. Then, retailer L can charge a relatively low price to attract consumers with moderate valuation. To retain these consumers, retailer H has an incentive to reduce the difference between his retail price and his competitor's retail price. As a result, the retail price of the high-quality product is lower, compared to the retail price in the scenario without return insurance. However, when the premium differentiation degree is relatively large (i.e., $(\frac{(2\gamma_H - 1)w}{\gamma_H} - \frac{3f_H}{\gamma_H - \gamma_L})^+ < \beta < \frac{(1-\gamma_L)w - f_H}{\gamma_H - \gamma_L}$), retailer H can also increase his price by a smaller extent than retailer L. Then, retail prices for both products are higher than those in the scenario without return insurance. Fig. 1 compares retail prices between the equilibrium case K^* and the case without offering return insurance $K = 00$, where the solid line represents the price in the equilibrium and the dotted line denotes the price in case $K = 00$. When the equilibrium $K^* = 00$, two lines overlap.

Proposition 3. *The adoption of return insurance enlarges the total market size (i.e., $D_L^{K^*} + D_H^{K^*} \geq D_L^{00} + D_H^{00}$, $K^* = 01$, 10, *or* 11). Moreover, regarding the demand of each retailer, we have the following results:*

1. When only one retailer offers return insurance (i.e., $K^* = 01$ or 10), the return insurance enables the retailer to increase his demand (i.e., $D_L^{10} \geq D_L^{00}$ and $D_H^{01} \geq D_H^{00}$) and his competitor obtains less demand (i.e., $D_H^{10} \leq D_H^{00}$ and $D_L^{01} \leq D_L^{00}$).

2. When both retailers offer return insurance (i.e., $K^* = 11$),

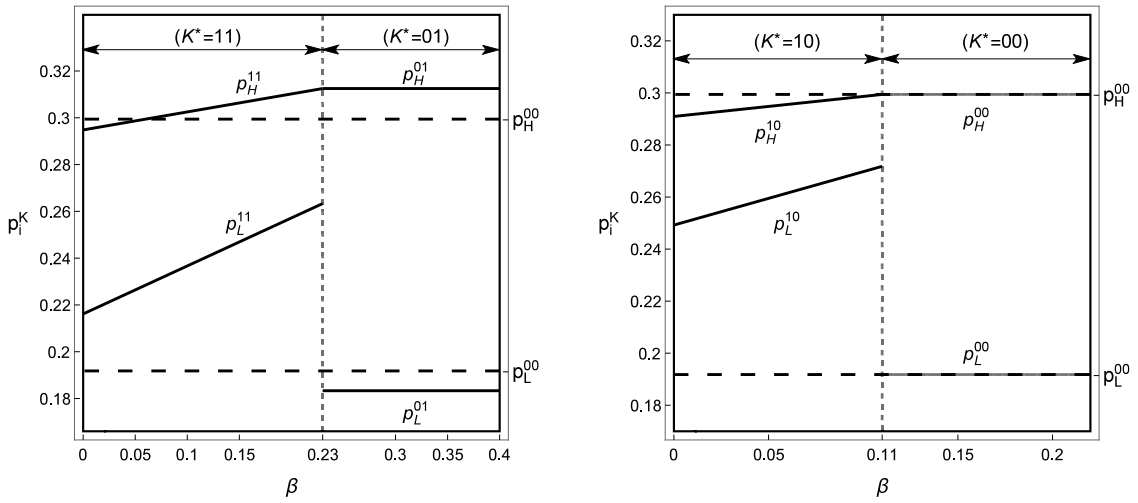
a. if $\beta \leq \frac{f_H + (2\gamma_H - 1)w}{\gamma_H}$, retailer L has more demand (i.e., $D_L^{11} > D_L^{00}$), while retailer H faces a reduction in demand (i.e., $D_H^{11} \leq D_H^{00}$);

b. if $\frac{f_H + (2\gamma_H - 1)w}{\gamma_H} < \beta < \frac{(2-\gamma_L)w - 2f_H}{2\gamma_H - \gamma_L}$, both retailer L and retailer H have more demand (i.e., $D_i^{11} > D_i^{00}$, $i \in \{L, H\}$);

c. if $\frac{(2-\gamma_L)w - 2f_H}{2\gamma_H - \gamma_L} \leq \beta < \frac{(1-\gamma_L)w - f_H}{\gamma_H - \gamma_L}$, retailer L faces a reduction in demand (i.e., $D_L^{11} \leq D_L^{00}$) and retailer H has more demand (i.e., $D_H^{11} > D_H^{00}$).

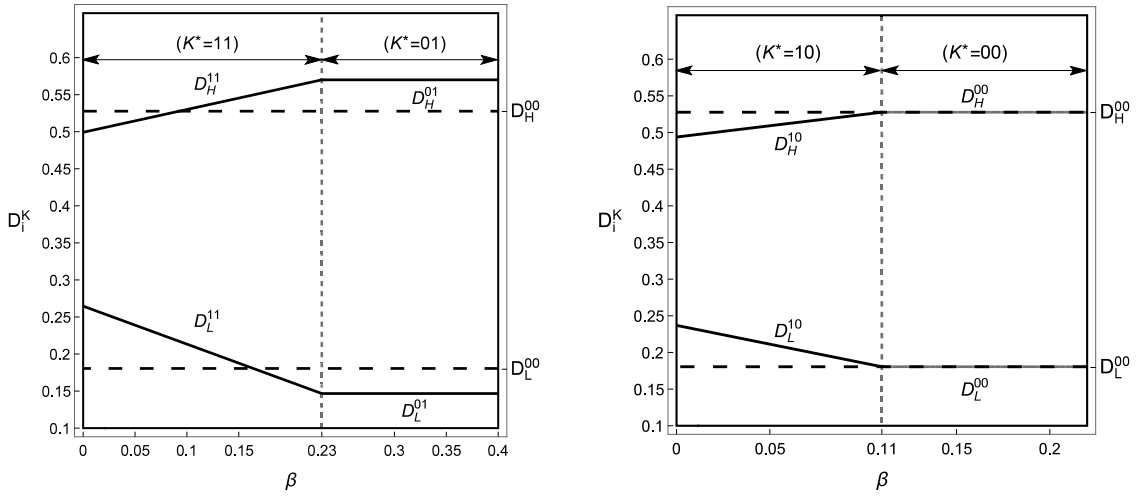
Proposition 3 indicates that the return insurance enables the retailers to serve consumers with low valuation and enlarge the total market size. Intuitively, the return insurance reduces the consumer's hassle cost and some consumers with low valuation are willing to buy a product. Thus, the total market size is enlarged. In addition, when only one retailer offers return insurance, the retailer can easily grab consumers from his competitor. Consequently, the competitor without return insurance faces a reduction in demand. Fig. 2 compares the demand in the equilibrium case with the case without return insurance, where the solid line represents the demand in equilibrium K^* and the dotted line denotes the demand in case $K = 00$. When the equilibrium $K^* = 00$, two lines overlap.

Differently, when the premium difference is moderate (i.e., $\frac{f_H + (2\gamma_H - 1)w}{\gamma_H} < \beta < \frac{(2-\gamma_L)w - 2f_H}{2\gamma_H - \gamma_L}$), both retailers L and H can have more demand after offering return insurance. In this case, retailer



(a) Low f_H: $f_H = 0.002$. (b) High f_H: $f_H = 0.026$.

Fig. 1. Retail prices in the equilibrium case K^* ($\gamma_H = 0.8$, $\gamma_L = 0.6$, $C_H = 0.15$, $C_L = 0.1$, $S_H - T_H = 0.08$, $S_L - T_L = 0.03$, $r = 0.15$, $w = 0.12$).



(a) Low f_H: $f_H = 0.002$. (b) High f_H: $f_H = 0.026$.

Fig. 2. Demand in the equilibrium case K^* ($\gamma_H = 0.8$, $\gamma_L = 0.6$, $C_H = 0.15$, $C_L = 0.1$, $S_H - T_H = 0.08$, $S_L - T_L = 0.03$, $r = 0.15$, $w = 0.12$).

L's cost in offering return insurance is moderate. As demonstrated in Proposition 2, retailer L intends to charge a relatively high retail price. Then, retailer H can also increase his price by a smaller extent than retailer L. Although both retailers increase the retail prices, the return insurance enables the consumers to obtain more utilities and more consumers purchase the products. However, if the premium differentiation degree is relatively large (i.e., $\beta \geq \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L}$), retailer L further increases his retail price by a sufficiently large extent, which leads more consumers to switch from product L to product H. Thus, retailer L faces a reduction in demand. In contrast, if the premium differentiation degree is small enough (i.e., $\beta \leq \frac{f_H+(2\gamma_H-1)w}{\gamma_H}$), the retailer L's cost in offering the return insurance is low. Retailer L would charge a relatively low price to grab the consumers, leading to a reduction in the demand for product H.

4.3. Impacts of the return insurance

In this section, we investigate the impacts of the return insurance on the retailers' profit, the consumer surplus, and the social welfare. We use CS^K and SW^K to denote the consumer surplus and the social welfare, where $K \in \{00, 01, 10, 11\}$. Following the study by Chen and Chen (2017), the consumer surplus CS^K and the social welfare SW^K are defined as follows:

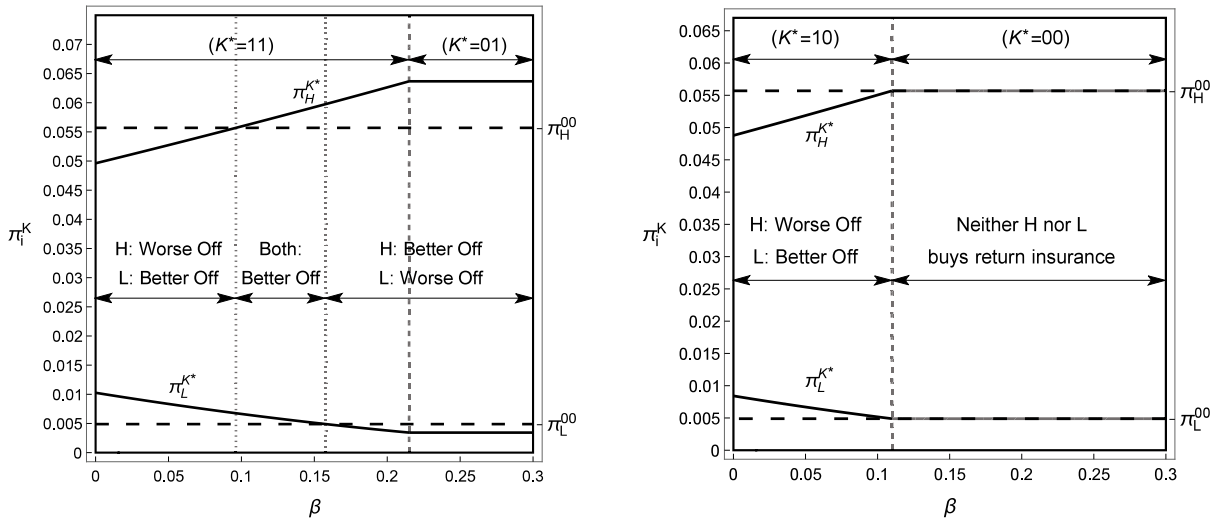
$$CS^K = \int_{v_L^K}^{v_{HL}^K} EU_L^K(v)dv + \int_{v_{HL}^K}^{1} EU_H^K(v)dv, \tag{8}$$

$$SW^K = CS^K + (\pi_L^K + \pi_H^K), \tag{9}$$

where $EU_L^K(v)$ ($EU_H^K(v)$) represents the surplus of a consumer with valuation v who purchases product L (H). In line with existing studies on return insurance (e.g., Li et al., 2018, 2021), our paper does not consider the profit of the insurer in social welfare.[10] Our results are summarized in the following propositions.

Proposition 4. *The impacts of return insurance on the retailers' profit are shown as follows:*

[10] If we consider the insurer's income, we have to analyze both the sales and the investment income of the premiums, which is very complicated and deviates from our focus in this paper.



(a) Low f_H: $f_H = 0.005$. (b) High f_H: $f_H = 0.026$.

Fig. 3. The impact of return insurance on two retailers' profit ($\gamma_H = 0.8$, $\gamma_L = 0.6$, $C_H = 0.15$, $C_L = 0.1$, $S_H - T_H = 0.08$, $S_L - T_L = 0.03$, $r = 0.15$, $w = 0.12$).

1. If $\beta \le \min\{\frac{f_H+(2\gamma_H-1)w}{\gamma_H}, \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}\}$,[11] *retailer L is better off (i.e.,* $\pi_L^{1k_H} \ge \pi_L^{00}$, $k_H \in \{0,1\}$*), while retailer H is worse off (i.e.,* $\pi_H^{1k_H} \le \pi_H^{00}, k_H \in \{0,1\}$*).*

2. If $f_H \le (1-\gamma_H)w$ *and* $\frac{f_H+(2\gamma_H-1)w}{\gamma_H} < \beta < \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L}$*, both retailers are better off (i.e.,* $\pi_i^{11} > \pi_i^{00}$, $i \in \{L, H\}$*), that is, the return insurance leads to the Pareto improvement.*

3. If $f_H \le (1-\gamma_H)w$ *and* $\beta > \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L}$*, retailer L is worse off (i.e.,* $\pi_L^{k_L 1} \le \pi_L^{00}$, $k_L \in \{0,1\}$*), while retailer H is better off (i.e.,* $\pi_H^{k_L 1} \ge \pi_H^{00}$, $k_L \in \{0,1\}$*).*

If the above conditions are not satisfied (i.e., $f_H > (1-\gamma_H)w$ *and* $\beta > \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$ *hold), neither retailer H nor retailer L purchases return insurance.*

Proposition 4 captures the impacts of the return strategy on the retailers' profit. Specifically, when the premium differentiation degree is sufficiently small (i.e., $\beta \le \min\{\frac{f_H+(2\gamma_H-1)w}{\gamma_H}, \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}\}$), retailer L offers return insurance regardless of retailer H's choice (i.e., $K^* = 10$ or 11). In this case, retailer L can gain more profit, while retailer H gains less profit. In contrast, when the premium for retailer H is low (i.e., $f_H \le (1-\gamma_H)w$) and the premium differentiation degree is large (i.e., $\beta > \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L}$), retailer H purchases return insurance regardless of retailer L's choice (i.e., $K^* = 01$ or 11). Then, retailer H can obtain more consumers and gain more profit, while retailer L earns less profit due to the demand reduction.

It is worth mentioning that when the premium for retailer H is low (i.e., $f_H \le (1-\gamma_H)w$) and the premium differentiation degree is moderate (i.e., $\frac{f_H+(2\gamma_H-1)w}{\gamma_H} < \beta < \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$), both retailers L and H can gain more profit by offering return insurance. In this case, return insurance is efficient in improving the consumers' willingness to pay, both retailers may charge higher retail prices and sell more products. Thus, the return insurance leads to the Pareto improvement. In Fig. 3(a), if the premium differentiation degree is moderate (e.g., $0.096 < \beta < 0.158$), both retailers purchase return insurance (i.e., $K^* = 11$), and a win-win outcome incurs. Fig. 3 compares profit between the equilibrium case K^* and $K = 00$, where the solid line represents the profit in equilibrium and the dotted line denotes the profit in case $K = 00$. When the equilibrium $K^* = 00$, two lines overlap.

Proposition 5. *Both the social welfare and the consumer surplus can be improved owing to the return insurance: (1)* $CS^{11} \ge CS^{01} \ge CS^{00}$ *and* $CS^{11} \ge CS^{10} \ge CS^{00}$*; (2) if* $\frac{\Delta_H^K}{\Delta_L^K} \in [\frac{\gamma_H(8\gamma_H-3\gamma_L)}{12(\gamma_H)^2-9\gamma_H\gamma_L+2(\gamma_L)^2}, \frac{12(\gamma_H)^2-9\gamma_H\gamma_L+2(\gamma_L)^2}{\gamma_L(8\gamma_H-3\gamma_L)}]$,[12] $SW^{11} \ge SW^{01} \ge SW^{00}$ *and* $SW^{11} \ge SW^{10} \ge SW^{00}$.

Proposition 5 demonstrates the impacts of return insurance on the consumer surplus and the social welfare. Specifically, as the retailer offers return insurance, the consumer surplus increases. Firstly, the adoption of return insurance can induce consumers with low valuation to buy, which enlarges the total market size, as shown in Proposition 3. Secondly, in the competitive market, the competition limits the extent of the price increase by the retailer who offers the return insurance, and forces his competitor to reduce the price. Note that under certain conditions as shown in Proposition 2, retailer H has to reduce the price to compete with retailer L when both retailers offer return insurance.

Moreover, when the difference between the retailers' *maximum net shared values* is moderate (i.e., $\frac{\Delta_H^K}{\Delta_L^K} \in [\frac{\gamma_H(8\gamma_H-3\gamma_L)}{12(\gamma_H)^2-9\gamma_H\gamma_L+2(\gamma_L)^2}, \frac{12(\gamma_H)^2-9\gamma_H\gamma_L+2(\gamma_L)^2}{\gamma_L(8\gamma_H-3\gamma_L)}]$), the return insurance improves the social welfare. When only one retailer offers the return insurance, the profit of the retailer can be improved, while the profit of his competitor reduces. However, if the difference between the retailers' *maximum net shared values* is moderate, the improvement of consumer surplus exceeds the loss of the total profit of two retailers, owing to the increase in the market size. Thus, the social welfare is improved. In addition, the social welfare reaches the maximum when both retailers offer return insurance. In this case, the consumer surplus is maximized and both retailers may obtain higher profit.

5. Discussion

In this section, we extend our basic model to the following scenarios: (1) online retailers endogenously determine the quality levels; (2) consumers' valuation is influenced by the quality. We would like to investigate the relationship between the return insurance and the retailers' quality decisions, and examine the robustness of our results in the basic model.

[11] Specifically, this condition is the same as condition set: $f_H \le (1-\gamma_H)w$ and $\beta \le \frac{f_H+(2\gamma_H-1)w}{\gamma_H}$ or condition set: $f_H > (1-\gamma_H)w$ and $\beta \le \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$.

[12] Note that $[\frac{\gamma_H(8\gamma_H-3\gamma_L)}{12(\gamma_H)^2-9\gamma_H\gamma_L+2(\gamma_L)^2}, \frac{12(\gamma_H)^2-9\gamma_H\gamma_L+2(\gamma_L)^2}{\gamma_L(8\gamma_H-3\gamma_L)}] \subset (\frac{\gamma_H}{2\gamma_H-\gamma_L}, \frac{2\gamma_H-\gamma_L}{\gamma_L})$, which is proved in the proof of *Proposition* 5 in *Appendix* A.

Table 3
Payoff matrix for two retailers ($\alpha = 0.45$, $S_i - T_i - r = 0.02$, $w = 0.015$, $\xi = 0.027$, and $f(\gamma_i) = \xi(1 - \sqrt{\gamma_i})$, where $i \in \{L, H\}$).

L	H	
	$k_H = 0$	$k_H = 1$
$k_L = 0$	0.028702, 0.039697	0.028647, 0.039787
$k_L = 1$	0.028585, 0.038915	0.028525, 0.039003

5.1. Online retailers endogenously determine the quality levels.

In this section, we consider the scenario wherein the retailers endogenously determine the quality levels, as in Huang et al. (2018). We assume that two retailers decide whether to offer the return insurance, their quality levels, and their retail prices. It is worth mentioning that the retailer's quality decision is reflected in his decision on the satisfaction rate. Specifically, given the retailer's optimal satisfaction rate γ_i^*, we can obtain that the retailer's optimal quality is $q_i^* = g^{-1}(\gamma_i^*)$. Besides, when the retailer chooses a higher satisfaction rate, he has to increase the quality. In the following, we characterize the retailers' quality decisions based on the satisfaction rates.

Moreover, we assume that retailer i incurs a product acquisition cost $\alpha\gamma_i^2$ to obtain the product with the satisfaction rate γ_i (i.e., $C_i = \alpha\gamma_i^2$), which is widely adopted by many studies (e.g., Chen and Chen, 2017; Huang et al., 2018). Following McWilliams (2012), we assume that the parameters S_i and T_i of two retailers are identical. Furthermore, the insurance premium $f_i = f(\gamma_i)$ and $f'(\cdot) < 0$, which means that the retailer should pay a higher return insurance premium for a lower satisfaction rate.

Following the relevant literature (e.g., Moorthy, 1988; Huang et al., 2018), we find that the two competing retailers are better off by differentiating their quality levels. If the quality levels of two retailers are identical, their satisfaction rates are also the same. Each retailer has an incentive to deviate to a lower price than his competitor to grab more market shares. As a result, both retailers charge identical prices which are equal to the marginal acquisition costs, and earn zero profit. For the sake of brevity, we relegate detailed analysis to Appendix B.1.

Then, we focus on two retailers' quality decisions and return insurance strategies when they differentiate their quality levels. Based on Lemma 1, we further examine two retailers' quality decisions in four cases (i.e., $K \in \{00, 01, 10, 11\}$). Then, we compare the profit and describe retailers' decisions on the return insurance. However, since the model is very complicated, we have to rely on numerical experiments to examine the interaction between the decisions on the quality levels and the return insurance. Moreover, following the study of Huang et al. (2018), we consider three types of insurance premium functions: (1) linear (i.e., $f(\gamma_i) = \xi(1 - \gamma_i)$); (2) convex (i.e., $f(\gamma_i) = \xi(1 - \sqrt{\gamma_i})$); (3) concave (i.e., $f(\gamma_i) = \xi(1 - \gamma_i^2)$).

We now analyze a numerical experiment as an example. We set $\alpha = 0.45$, $S_i - T_i - r = 0.02$, $w = 0.015$, $\xi = 0.027$, $f(\gamma_i) = \xi(1 - \sqrt{\gamma_i})$, and $i \in \{L, H\}$. In case $K = 01$, we obtain that the retailers' optimal prices $p_L^{01}(\gamma_H, \gamma_L)$ and $p_H^{01}(\gamma_H, \gamma_L)$ and corresponding profit $\pi_L^{01}(\gamma_H, \gamma_L)$ and $\pi_H^{01}(\gamma_H, \gamma_L)$, as in Lemma 1. From the first-order conditions, $\frac{\partial \pi_L^{01}(\gamma_H, \gamma_L)}{\partial \gamma_L} = 0$ and $\frac{\partial \pi_H^{01}(\gamma_H, \gamma_L)}{\partial \gamma_H} = 0$. We use Matlab to solve these two equations and obtain that $\gamma_L^{01} = 0.3920$ and $\gamma_H^{01} = 0.8673$. Then, the retailers' optimal quality levels are $q_L^{01} = g^{-1}(0.3920)$ and $q_H^{01} = g^{-1}(0.8673)$, and the retailers' payoffs are $\pi_L^{01} = 0.028647$ and $\pi_H^{01} = 0.039787$. Moreover, following the same procedure, we obtain retailers' decisions on quality levels in the other three cases. Specifically, given $K = 00$, $q_L^{00} = g^{-1}(0.3926)$ and $q_H^{00} = g^{-1}(0.8684)$; given $K = 10$, $q_L^{10} = g^{-1}(0.4027)$ and $q_H^{10} = g^{-1}(0.8730)$; given $K = 11$, $q_L^{11} = g^{-1}(0.4021)$ and $q_H^{11} = g^{-1}(0.8718)$. As a result, the payoff matrix for two retailers is shown in Table 3.

From Table 3, we find that $K = 01$ is the Nash equilibrium (i.e., $K^* = 01$). Specifically, regardless of retailer L's choice, offering return insurance (i.e., $k_H = 1$) is the dominant strategy for retailer H since $\pi_H^{01} = 0.039787 > \pi_H^{00} = 0.039697$ and $\pi_H^{11} = 0.039003 > \pi_H^{10} = 0.038915$. Similarly, regardless of retailer H's choice, non-adoption of the return insurance (i.e., $k_L = 0$) is the dominant strategy for retailer L since $\pi_L^{00} = 0.028702 > \pi_L^{10} = 0.028585$ and $\pi_L^{01} = 0.028647 > \pi_L^{11} = 0.028525$. Therefore, retailer L abandons the return insurance and retailer H offers the return insurance. Following the same procedure, we derive the retailers' optimal quality levels and characterize the equilibrium strategies through several numerical experiments. For the sake of brevity, we present a part of the numerical results in Tables 4 and 5 and relegate the detailed results to Appendix B.1.

Moreover, we note that if $(1 - \gamma_i)w - f(\gamma_i) \geq 0$, we call that offering return insurance is an efficient strategy, which indicates that the return insurance provides enough value for a retailer or a consumer owing to a high shipping compensation or a low premium. Then, as ξ/w decreases, the return insurance is more likely to be efficient for retailers. We present some results in the following observations, which are derived from the numerical experiments shown in Tables 4 and 5. For simplicity, we do not provide the retailers' optimal quality levels in Tables 4 and 5, and note that the increase or decrease in the quality is the same as that in the satisfaction rate.

Observation 1. *When ξ/w is not high, both retailers purchase return insurance for consumers (i.e., $K^* = 11$). Concerning the quality decisions, we have the following results: (1) if ξ/w is moderate, retailers L and H successively improve their quality levels as ξ/w increases when $f(\gamma_i) = \xi(1 - \sqrt{\gamma_i})$; (2) otherwise,*[13] *both retailers reduce the quality levels.*

Observation 1 identifies the condition under which both retailers purchase return insurance for consumers. A lower ξ/w indicates that the return insurance is more likely to be efficient for the retailers. Then, both retailers choose to purchase return insurance when ξ/w is not high. When both retailers purchase return insurance and the premium function is convex, the retailers may improve their quality levels (satisfaction rates). Specifically, when the premium function is convex, a retailer with higher quality is more efficient to offer return insurance. If ξ/w increases, retailer L firstly improves the quality. Then, the satisfaction rate is higher and more consumers are willing to buy product L. Facing retailer L's quality adjustment, retailer H has to increase the quality to reduce the premium cost and grab more market shares, even though the additional acquisition cost is high. Note that retailer L's additional cost of improving the quality is lower than that of retailer H and thus retailer L is willing to improve the quality before retailer H. Except the case aforementioned, we find that both retailers reduce the quality levels when offering return insurance. In these cases, the insurance can reduce the consumers' hassle cost and the consumers are less sensitive to the satisfaction rate. Then, retailers have an incentive to reduce the quality levels to earn more profit by cutting the acquisition costs, resulting in lower satisfaction rates.

Observation 2. *When both retailers L and H improve the quality levels in the equilibrium $K^* = 11$, both retailers face a reduction in their profit. However, both the consumer surplus and the social welfare increase.*

Observation 2 demonstrates that when both retailers purchase return insurance and improve quality levels, both retailers face a reduction in their profit. Intuitively, higher quality leads to a higher product acquisition cost. Then, the retailer would charge a higher retail price, which causes a demand reduction. Moreover, recall that ξ/w is relatively low in this case as shown in Observation 1 and the increase in the retail price is relatively small. Then, the price adjustment cannot

[13] Both retailers purchase return insurance and reduce the quality levels when one of the following condition sets is satisfied: (1) $f(\gamma_i) = \xi(1 - \sqrt{\gamma_i})$ and ξ/w is sufficiently low; (2) $f(\gamma_i) = \xi(1 - \gamma_i^2)$ and ξ/w is not high; (3) $f(\gamma_i) = \xi(1 - \gamma_i)$ and ξ/w is not high.

Table 4
Equilibria when $f(\gamma_i) = \xi(1-\sqrt{\gamma_i})$ ($\alpha = 0.45$, $S_i - T_i - r = 0.02$, $w = 0.015$, where $i \in \{L, H\}$).

ξ/w	Equilibrium	γ_L	γ_H	p_L	p_H	D_L	D_H	CS	SW
0.8	$K^* = 11$	0.3737	0.8510	0.239978	0.532692	0.374716	0.293129	0.103844	0.174286
1	$K^* = 11$	0.3797	0.8533	0.245457	0.532312	0.371984	0.292708	0.104267	0.173905
1.3	$K^* = 11$	0.3884	0.8617	0.254366	0.534534	0.368474	0.290554	0.104323	0.173245
1.5	$K^* = 11$	0.3940	0.8658	0.259666	0.535267	0.366160	0.289579	0.104491	0.172840
1.55	$K^* = 11$	0.3954	0.8669	0.260987	0.535490	0.365602	0.289312	0.104528	0.172739
1.6	$K^* = 11$	0.3967	0.8678	0.262206	0.535640	0.365041	0.289109	0.104565	0.172638
1.65	$K^* = 11$	0.3981	0.8689	0.263510	0.535862	0.364493	0.288841	0.104603	0.172539
1.7	$K^* = 11$	0.3994	0.8698	0.264714	0.536012	0.363942	0.288638	0.104639	0.172439
1.75	$K^* = 01$	0.3916	0.8666	0.232653	0.536943	0.365123	0.289598	0.103850	0.172302
1.8	$K^* = 01$	0.3920	0.8673	0.236931	0.537210	0.365173	0.289325	0.103853	0.172287
1.85	$K^* = 01$	0.3923	0.8680	0.237280	0.537512	0.365230	0.289055	0.103843	0.172268
1.9	$K^* = 00$	0.3926	0.8684	0.237570	0.535402	0.365285	0.288847	0.103843	0.172243

Table 5
Equilibria when $f(\gamma_i) = \xi(1-\gamma_i^2)$ ($\alpha = 0.45$, $S_i - T_i - r = 0.02$, $w = 0.015$, where $i \in \{L, H\}$).

ξ/w	Equilibrium	γ_L	γ_H	p_L	p_H	D_L	D_H	CS	SW
0.4	$K^* = 11$	0.3692	0.8535	0.239086	0.536097	0.374815	0.292128	0.102777	0.173538
0.5	$K^* = 11$	0.3745	0.8588	0.245264	0.537925	0.372108	0.290766	0.102750	0.172938
0.55	$K^* = 11$	0.3771	0.8614	0.248261	0.538725	0.370783	0.290106	0.102734	0.172641
0.6	$K^* = 10$	0.3789	0.8626	0.250412	0.536514	0.369319	0.290112	0.102737	0.172428
0.65	$K^* = 10$	0.3802	0.8627	0.252055	0.536319	0.367869	0.290418	0.102725	0.172197
0.7	$K^* = 00$	0.3926	0.8684	0.23757	0.535402	0.365285	0.288847	0.103843	0.172243

counterbalance the loss caused by the demand reduction and the product acquisition cost increase. Therefore, both retailers face a reduction in their profit.

However, both the consumer surplus and the social welfare increase. When retailers increase their quality levels to increase the satisfaction rates, consumers obtain more gross utilities. Moreover, due to the competition between retailers, the retailers are unable to increase the retail prices by a large extent. Therefore, the consumer surplus is increased. Furthermore, the increase in the consumer surplus exceeds the reduction in the total profit of both retailers. Thus, the social welfare is improved.

Observation 3. *The equilibria in which only one retailer purchases return insurance occurs in the following cases:*

1. When ξ/w is high and $f(\gamma_i) = \xi(1-\sqrt{\gamma_i})$, only retailer H purchases return insurance (i.e., $K^* = 01$). In this case, both retailers reduce the quality levels. Moreover, retailer L reduces the retail price and earns less profit, while retailer H increases the retail price and earns more profit.

2. When ξ/w is relatively high and $f(\gamma_i) = \xi(1-\gamma_i^2)$, only retailer L purchases return insurance (i.e., $K^* = 10$). In this case, both retailers reduce the quality levels, increase retail prices, and earn more profit.

Observation 3 identifies the conditions under which only one retailer purchases return insurance. Firstly, when the premium function is convex and ξ/w is high, only retailer H purchases return insurance. As in Observation 1, when the premium function is convex, the retailer with higher quality is more efficient to offer return insurance. When ξ/w is high, retailer H enjoys a large advantage in premiums, which enables him easily to grab consumers from retailer L and earn more profit by offering return insurance. However, for retailer L, the premium is costly and the additional profit brought by the insurance cannot counterbalance the cost of premiums. Thus, retailer L gives up buying return insurance. Consequently, retailer H earns more profit with more demand and a higher retail price, and retailer L without insurance has to reduce the retail price, and earns less profit due to the reduction in demand and retail price.

Secondly, when the premium function is concave and ξ/w is relatively high, only retailer L purchases return insurance. In contrast with the convex premium function, the concave premium function enables the retailer with lower quality to be more efficient to offer return insurance. Then, we can find that only retailer L purchases return insurance for consumers, both retailers reduce the quality levels (satisfaction rates) to cut product acquisition costs and earn more profit. Moreover, retailer L charges a higher retail price for more profit, and retailer H can also increase the retail price by a smaller extent to earn more profit.

5.2. Consumers' valuation is influenced by the quality.

In this section, we consider the case where both the satisfaction rates and the consumers' valuation are influenced by the quality. Specifically, when consumers are satisfied with product i ($i \in \{L, H\}$), their valuation is θq_i. Along with the study of Guo (2009), θ captures the heterogeneity in consumers' willingness to pay for quality and θ is uniformly distributed between 0 and 1 (i.e., $\theta \sim U[0,1]$). Moreover, the satisfaction rate γ_i increases in the quality level, that is, $\gamma_i = g(q_i)$ and $g'(\cdot) > 0$. Note that the other assumptions remain the same as those in the basic model in Section 3.

Then, we obtain that the consumers' expected utility of purchasing a product from retailer i is

$$EU_i^K = \gamma_i(\theta q_i - p_i^K) - (1-\gamma_i)(r - k_i w). \tag{10}$$

Following the same procedure in the basic model, we characterize the retailers' pricing strategies in four cases (i.e., $K \in \{00, 01, 10, 11\}$). As in Section 4.1, we define $\hat{\Delta}_i^K = \gamma_i q_i - C_i + (1-\gamma_i)(S_i - T_i - r + k_i w) - k_i f_i$ as the *maximum net shared value* of retailer i, where $i = \{L, H\}$. In the following, we present the retailers' pricing strategies, demand, and profit in Lemma 2 and the values of $\hat{\Delta}_i^K$ are summarized in Table 6.

Lemma 2. *When the consumers' valuation is influenced by the quality, if $\frac{\gamma_H q_H}{2\gamma_H q_H - \gamma_L q_L} < \frac{\hat{\Delta}_H^K}{\hat{\Delta}_L^K} < \frac{2\gamma_H q_H - \gamma_L q_L}{\gamma_L q_L}$, the retailers' optimal retail prices in case K are*

$$\begin{cases} \hat{p}_L^K = q_L - \frac{\gamma_L q_L \hat{\Delta}_H^K + 2\gamma_H q_H \hat{\Delta}_L^K}{\gamma_L(4\gamma_H q_H - \gamma_L q_L)} - \frac{(1-\gamma_L)(r-k_L w)}{\gamma_L}, \\ \hat{p}_H^K = q_H[1 - \frac{2\hat{\Delta}_H^K + \hat{\Delta}_L^K}{4\gamma_H q_H - \gamma_L q_L}] - \frac{(1-\gamma_H)(r-k_H w)}{\gamma_H}. \end{cases} \tag{11}$$

Moreover, their demand and profit are shown as follows:

$$\begin{aligned} \hat{D}_L^K &= \frac{\gamma_H q_H[(2\gamma_H q_H - \gamma_L q_L)\hat{\Delta}_L^K - \gamma_L q_L \hat{\Delta}_H^K]}{\gamma_L q_L(4\gamma_H q_H - \gamma_L q_L)(\gamma_H q_H - \gamma_L q_L)}, \\ \hat{D}_H^K &= \frac{(2\gamma_H q_H - \gamma_L q_L)\hat{\Delta}_H^K - \gamma_H q_H \hat{\Delta}_L^K}{(4\gamma_H q_H - \gamma_L q_L)(\gamma_H q_H - \gamma_L q_L)}, \end{aligned} \tag{12}$$

Table 6
The values of $\hat{\Delta}_i^K$, $i = \{L, H\}$ and $K = \{00, 01, 10, 11\}$.

	$\hat{\Delta}_L^K$	$\hat{\Delta}_H^K$
$K = 00$	$\gamma_L q_L - C_L + (1-\gamma_L)(S_L - T_L - r)$	$\gamma_H q_H - C_H + (1-\gamma_H)(S_H - T_H - r)$
$K = 01$	$\gamma_L q_L - C_L + (1-\gamma_L)(S_L - T_L - r)$	$\gamma_H q_H - C_H + (1-\gamma_H)(S_H - T_H - r + w) - f_H$
$K = 10$	$\gamma_L q_L - C_L + (1-\gamma_L)(S_L - T_L - r + w) - f_L$	$\gamma_H q_H - C_H + (1-\gamma_H)(S_H - T_H - r)$
$K = 11$	$\gamma_L q_L - C_L + (1-\gamma_L)(S_L - T_L - r + w) - f_L$	$\gamma_H q_H - C_H + (1-\gamma_H)(S_H - T_H - r + w) - f_H$

$$\hat{\pi}_L^K = \frac{\gamma_H q_H[(2\gamma_H q_H - \gamma_L q_L)\hat{\Delta}_L^K - \gamma_L q_L \hat{\Delta}_H^K]^2}{\gamma_L q_L(4\gamma_H q_H - \gamma_L q_L)^2(\gamma_H q_H - \gamma_L q_L)},$$
$$\hat{\pi}_H^K = \frac{[(2\gamma_H q_H - \gamma_L q_L)\hat{\Delta}_H^K - \gamma_H q_H \hat{\Delta}_L^K]^2}{(4\gamma_H q_H - \gamma_L q_L)^2(\gamma_H q_H - \gamma_L q_L)}, \quad (13)$$

where $K \in \{00, 01, 10, 11\}$.

In Lemma 2, the condition $\frac{\gamma_H q_H}{2\gamma_H q_H - \gamma_L q_L} < \frac{\hat{\Delta}_H^K}{\hat{\Delta}_L^K} < \frac{2\gamma_H q_H - \gamma_L q_L}{\gamma_L q_L}$ ensures that two competing retailers co-exist.

Lemma 3. *When consumers' valuation is influenced by the quality, retailer* i *(*$i \in \{L, H\}$*) purchases return insurance for consumers (i.e.,* $k_i = 1$*) if and only if* $(1-\gamma_i)w - f_i \geq 0$*, independent of the competitor's decision.*

Lemma 3 indicates that two retailers' return insurance strategies are the same as those in the basic model, which verifies the robustness of our findings in Proposition 1.

Proposition 6. *When consumers' valuation is influenced by the quality, the return insurance's impacts on the retailers' profit are shown as follows:*

1. If $\beta \leq \min\{\underline{\beta}, \frac{(1-\gamma_L)w - f_H}{\gamma_H - \gamma_L}\}$*, retailer L is better off (i.e.,* $\hat{\pi}_L^{1k_H} \geq \hat{\pi}_L^{00}$*), while retailer H is worse off (i.e.,* $\hat{\pi}_H^{1k_H} \leq \hat{\pi}_H^{00}$*), where* $k_H \in \{0, 1\}$ *and* $\underline{\beta} = w - \frac{(1-\gamma_H)w - f_H}{\gamma_H - \gamma_L}[1 - \frac{\gamma_L}{\gamma_H \cdot (\frac{q_H}{q_L})}]$.

2. If $f_H \leq (1-\gamma_H)w$ *and* $\underline{\beta} < \beta < \overline{\beta}$*, both retailers are better off (i.e.,* $\hat{\pi}_L^{11} > \hat{\pi}_L^{00}$ *and* $\hat{\pi}_H^{11} > \hat{\pi}_H^{00}$*), that is, the return insurance leads to the Pareto improvement, where* $\overline{\beta} = w + \frac{(1-\gamma_H)w - f_H}{\gamma_H - \gamma_L}[1 - \frac{\gamma_L}{2\gamma_H \cdot (\frac{q_H}{q_L}) - \gamma_L}]$.

3. If $f_H \leq (1-\gamma_H)w$ *and* $\beta > \overline{\beta}$*, retailer L is worse off (i.e.,* $\hat{\pi}_L^{k_L 1} \leq \hat{\pi}_L^{00}$*), while retailer H is better off (i.e.,* $\hat{\pi}_H^{k_L 1} \geq \hat{\pi}_H^{00}$*), where* $k_L \in \{0, 1\}$.

If the above conditions are not satisfied (i.e., $f_H > (1-\gamma_H)w$ *and* $\beta > \frac{(1-\gamma_L)w - f_H}{\gamma_H - \gamma_L}$ *hold), neither retailer H nor retailer L purchases return insurance. Note that since* $q_H > q_L$*,* $\underline{\beta} < \frac{f_H + (2\gamma_H - 1)w}{\gamma_H}$ *and* $\overline{\beta} > \frac{(2-\gamma_L)w - 2f_H}{2\gamma_H - \gamma_L}$.

Proposition 6 demonstrates that the impacts of the return insurance on the retailers' profit are similar to those in the basic model (Proposition 4). Compared to the basic model, the range in which the return insurance enables two retailers to increase their demand is larger. Consumer heterogeneity increases the efficiency realized from purchasing return insurance. Thus, both retailers can have more demand and benefit from offering return insurance. In Proposition 4, if $\frac{q_H}{q_L} = 1$, $\overline{\beta} = \frac{(2-\gamma_L)w - 2f_H}{2\gamma_H - \gamma_L} = 0.096$ and $\underline{\beta} = \frac{f_H + (2\gamma_H - 1)w}{\gamma_H} = 0.158$, which are the upper and lower bounds for the emergence of Pareto improvement. However, in Fig. 4, given $\frac{q_H}{q_L} = 2.5$, both retailers purchase return insurance and a win-win outcome incurs if $0.053 < \beta < 0.197$, where $0.053 < 0.096$ and $0.197 > 0.158$, which indicates a larger range of β than the range in the basic model.

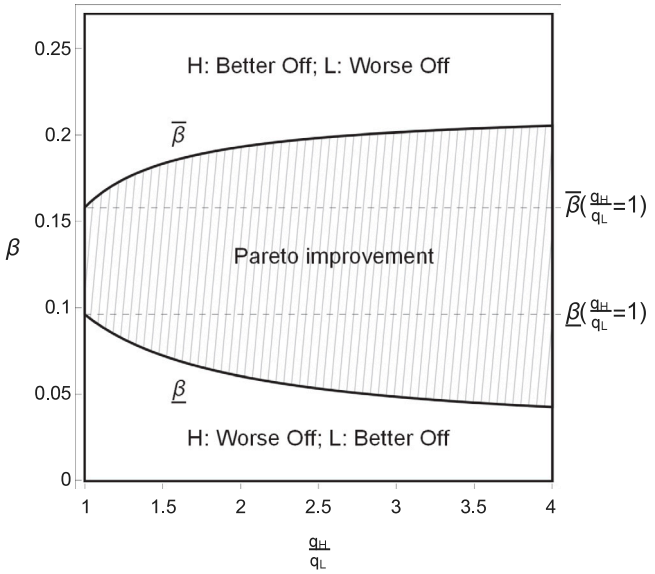


Fig. 4. The impact of return insurance on two retailers' profit ($\gamma_H = 0.8$, $\gamma_L = 0.6$, $f_H = 0.005$, $w = 0.12$).

6. Conclusions

In this paper, we study two competing online retailers' decisions on whether to purchase return insurance for consumers who have uncertainty about their satisfaction. Our study provides several managerial implications. Firstly, we suggest that when the premium for the high-quality retailer is sufficiently low and the premium for the low-quality retailer is sufficiently high, only the high-quality retailer should purchase return insurance for consumers. The low-quality retailer abandons the return insurance due to the high premium. In contrast, when the premiums for both retailers are high and the premium for the low-quality retailer is lower than or equal to the threshold, only the low-quality retailer purchases return insurance. Our result may provide an explanation for the practice in the cellphone retailing: Apple, as a high-quality retailer, does not offer return insurance, while his competitor with lower quality purchases return insurance.

Secondly, we suggest that when the premiums for the high-quality retailer and the low-quality retailer are low, both retailers offer return insurance. However, these retailers who purchase return insurance (e.g., Xiaomi and KONKA) should not blindly increase their prices. If the premium differentiation degree is sufficiently small, the high-quality retailer should reduce the retail price. In addition, we find that when both retailers offer return insurance, they can have more demand if the premium differentiation degree is moderate. Although both retailers increase the retail prices, the return insurance enables the consumers to obtain more utilities and more consumers purchase the products.

Thirdly, when the retailers endogenously decide their quality levels, both retailers may improve the quality levels if the premium function is convex. In this case, both retailers intend to reduce the premium by improving the quality, which can enhance the efficiency of the return insurance. Accordingly, our paper suggests that the insurance providers can adjust the insurance premium scheme to encourage the retailers to increase their quality levels.

There are several limitations to our work. Firstly, we do not consider the scenario in which consumers can buy the insurance. In this case, consumers are heterogeneous on the valuation of premium, which depends on the consumers' return rates on the online platform. This case is intriguing and deserves to be examined in future research. Secondly, the insurance premium and compensation can be endogenously determined by the third-party insurance provider. These decisions would potentially lead to more interesting interactions between competing retailers, and are remained to be investigated in future studies.

Data availability

No data was used for the research described in the article.

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Appendix

In the appendix, we first provide a detailed analysis of the setting of exogenous quality levels. Then, we extend our basic model, discuss case when retailers endogenously determine the quality levels, and the case when consumers' valuation is influenced by the quality.

Appendix A. Proof of main results.

Proof of Lemma 1. Firstly, we consider the case in which $K = 01$ as an example. In this case, retailer L gives up offering the return insurance (i.e., $k_L = 0$) and retailer H offers the return insurance (i.e., $k_H = 1$). At stage 2, based on Eq. (2), given $\underline{p}_H^{01} < p_H < \overline{p}_H^{01}$, we have the demand for retailer L and retailer H in the case of $K = 01$ as follows:

$$D_L^{01} = v_{HL}^{01} - v_L^{01}, \text{ and } D_H^{01} = 1 - v_{HL}^{01}. \tag{14}$$

where $v_L^{01} = [\gamma_L p_L + (1-\gamma_L)r]/\gamma_L$, $v_{HL}^{01} = [\gamma_H p_H - \gamma_L p_L + (1-\gamma_H)(r-w) - (1-\gamma_L)r]/(\gamma_H - \gamma_L)$, $v_H^{01} = [\gamma_H p_H + (1-\gamma_H)(r-w)]/\gamma_H$, $\overline{p}_H^{01} = [\gamma_H - \gamma_L + \gamma_L p_L - (1-\gamma_H)w + (\gamma_H - \gamma_L)r]/\gamma_H$, $\underline{p}_H^{01} = p_L + [(\gamma_H - \gamma_L)r + \gamma_L(1-\gamma_H)w]/\gamma_H\gamma_L$.

At stage 1, both online retailers decide on the retail prices, and their optimization problems are written as follows.

$$\max_{p_L^{01}} \pi_L^{01} = D_L^{01}[\gamma_L p_L^{01} + (1-\gamma_L)(S_L - T_L) - C_L], \text{ and} \tag{15}$$

$$\max_{p_H^{01}} \pi_H^{01} = D_H^{01}[\gamma_H p_H^{01} + (1-\gamma_H)(S_H - T_H) - C_H - f_H]. \tag{16}$$

From the first-order conditions, we have

$$2\gamma_H p_H^{01} - \gamma_L p_L^{01} - 2\gamma_H + \gamma_L - 2(1-\gamma_H)w + \Delta_H^{01} + (1 - 2\gamma_H + \gamma_L)r = 0, \text{ and} \tag{17}$$

$$\gamma_H\gamma_L p_H^{01} - 2\gamma_H\gamma_L p_L^{01} + \gamma_H\gamma_L - \gamma_L(1-\gamma_H)w + \gamma_H\Delta_L^{01} + (\gamma_H\gamma_L - 2\gamma_H + \gamma_L)r = 0. \tag{18}$$

Then, we can obtain that $p_L^{01} = 1 - \frac{\gamma_L\Delta_H^{01} + 2\gamma_H\Delta_L^{01}}{\gamma_L(4\gamma_H - \gamma_L)} - \frac{(1-\gamma_L)r}{\gamma_L}$ and $p_H^{01} = 1 - \frac{2\Delta_H^{01} + \Delta_L^{01}}{4\gamma_H - \gamma_L} - \frac{(1-\gamma_H)(r-w)}{\gamma_H}$, where $\Delta_H^{01} = \gamma_H - C_H + (1-\gamma_H)(S_H - T_H - r + w) - f_H$ and $\Delta_L^{01} = \gamma_L - C_L + (1-\gamma_L)(S_L - T_L - r)$. After substituting the price variables into Eq. (14), we have $D_L^{01} = \frac{\gamma_H[(2\gamma_H - \gamma_L)\Delta_L^{01} - \gamma_L\Delta_H^{01}]}{\gamma_L(4\gamma_H - \gamma_L)(\gamma_H - \gamma_L)}$ and $D_H^{01} = \frac{(2\gamma_H - \gamma_L)\Delta_H^{01} - \gamma_H\Delta_L^{01}}{(4\gamma_H - \gamma_L)(\gamma_H - \gamma_L)}$. Thus, we can obtain the optimal profit as follows: $\pi_L^{01} = \frac{\gamma_H[(2\gamma_H - \gamma_L)\Delta_L^{01} - \gamma_L\Delta_H^{01}]^2}{\gamma_L(4\gamma_H - \gamma_L)^2(\gamma_H - \gamma_L)}$, and $\pi_H^{01} = \frac{[(2\gamma_H - \gamma_L)\Delta_H^{01} - \gamma_H\Delta_L^{01}]^2}{(4\gamma_H - \gamma_L)^2(\gamma_H - \gamma_L)}$. Following the same procedure, we drive the solutions of the online retailers' optimal pricing strategies in other three cases (i.e., $K = 00, 10$, or 11) and show these results in Lemma 1.

Moreover, from Eq. (6) in Lemma 1, we can find that both retailers have positive sales (co-exist) if and only if $D_L^K > 0$ and $D_H^K > 0$. From $D_L^K > 0$, we obtain that $\frac{\Delta_H^K}{\Delta_L^K} < \frac{2\gamma_H - \gamma_L}{\gamma_L}$. From $D_H^K > 0$, $\frac{\Delta_H^K}{\Delta_L^K} > \frac{\gamma_H}{2\gamma_H - \gamma_L}$. To sum up, we have

(1) if $\frac{\Delta_H^K}{\Delta_L^K} \le \frac{\gamma_H}{2\gamma_H - \gamma_L}$, retailer H has no sales;

(2) if $\frac{\gamma_H}{2\gamma_H - \gamma_L} < \frac{\Delta_H^K}{\Delta_L^K} < \frac{2\gamma_H - \gamma_L}{\gamma_L}$, both retailers H and L have positive sales;

(3) if $\frac{\Delta_H^K}{\Delta_L^K} \ge \frac{2\gamma_H - \gamma_L}{\gamma_L}$, retailer L has no sales.

Proof of Proposition 1. From Table 2, we can easily find that $\Delta_L^{k_L 0} = \Delta_L^{k_L 1}$, $\Delta_H^{0k_H} = \Delta_H^{1k_H}$, $\Delta_L^{1k_H} - \Delta_L^{0k_H} = (1-\gamma_L)w - f_L$, and $\Delta_H^{k_L 1} - \Delta_H^{k_L 0} = (1-\gamma_H)w - f_H$, where $k_L \in \{0,1\}$ and $k_H \in \{0,1\}$.

We have

$$\pi_L^{1k_H} - \pi_L^{0k_H} = \frac{[(1-\gamma_L)w - f_L]\cdot\gamma_H(2\gamma_H - \gamma_L)\cdot A_1}{\gamma_L(4\gamma_H - \gamma_L)^2(\gamma_H - \gamma_L)}, \text{ and} \tag{19}$$

$$\pi_H^{k_L 1} - \pi_H^{k_L 0} = \frac{[(1-\gamma_H)w - f_H]\cdot(2\gamma_H - \gamma_L)\cdot A_2}{(4\gamma_H - \gamma_L)^2(\gamma_H - \gamma_L)}, \tag{20}$$

where $A_1 = (2\gamma_H - \gamma_L)(\Delta_L^{1k_H} + \Delta_L^{0k_H}) - \gamma_L(\Delta_H^{1k_H} + \Delta_H^{0k_H})$, and $A_2 = (2\gamma_H - \gamma_L)(\Delta_H^{k_L 1} + \Delta_H^{k_L 0}) - \gamma_H(\Delta_L^{k_L 1} + \Delta_L^{k_L 0})$. Given the condition that $\frac{\gamma_H}{2\gamma_H - \gamma_L} < \frac{\Delta_H^K}{\Delta_L^K} < \frac{2\gamma_H - \gamma_L}{\gamma_L}$, we can easily find that $A_1 > 0$ and $A_2 > 0$, $\pi_L^{1k_H} - \pi_L^{0k_H} \ge 0$ sustains if $(1-\gamma_L)w - f_L \ge 0$, and $\pi_H^{k_L 1} - \pi_H^{k_L 0} \ge 0$ sustains if $(1-\gamma_H)w - f_H \ge 0$.

Following the study by Zhang et al. (2022), we assume that when there is no difference between the online retailer on whether to purchase return insurance or not, the retailer always adopts the beneficial option for consumers, that is purchasing return insurance for consumers. Then, we have

Case (1): $K = 00$ emerges as the market equilibrium when the following conditions are satisfied: (1-a) $\pi_L^{00} > \pi_L^{10}$; (1-b) $\pi_H^{00} > \pi_H^{01}$.

Case (2): $K = 01$ emerges as the market equilibrium when the following conditions are satisfied: (2-a) $\pi_L^{01} > \pi_L^{11}$; (2-b) $\pi_H^{01} \ge \pi_H^{00}$.

Case (3): $K = 10$ emerges as the market equilibrium when the following conditions are satisfied: (3-a) $\pi_L^{10} \ge \pi_L^{00}$; (3-b) $\pi_H^{10} > \pi_H^{11}$.

Case (4): $K = 11$ emerges as the market equilibrium when the following conditions are satisfied: (4-a) $\pi_L^{11} \ge \pi_L^{01}$; (4-b) $\pi_H^{11} \ge \pi_H^{10}$.

Recall that $\beta = \frac{f_L - f_H}{\gamma_H - \gamma_L}$, we denote $f_L = f_H + \beta(\gamma_H - \gamma_L)$. Then, we can update the conditions of the equilibrium case. Specifically,

Case (1): $K^* = 00$: (1-a) $f_L > (1-\gamma_L)w$ (i.e., $\beta > \frac{(1-\gamma_L)w - f_H}{\gamma_H - \gamma_L}$); (1-b) $f_H > (1-\gamma_H)w$.

Case (2): $K^* = 01$: (2-a) $f_L > (1-\gamma_L)w$ (i.e., $\beta > \frac{(1-\gamma_L)w - f_H}{\gamma_H - \gamma_L}$); (2-b) $f_H \le (1-\gamma_H)w$.

Case (3): $K^* = 10$: (3-a) $f_L \le (1-\gamma_L)w$ (i.e., $\beta \le \frac{(1-\gamma_L)w - f_H}{\gamma_H - \gamma_L}$); (3-b) $f_H > (1-\gamma_H)w$.

Case (4): $K^* = 11$: (4-a) $f_L \le (1-\gamma_L)w$ (i.e., $\beta \le \frac{(1-\gamma_L)w - f_H}{\gamma_H - \gamma_L}$); (4-b) $f_H \le (1-\gamma_H)w$.

Proof of Proposition 2. Given the optimal prices in Eqs. (5), we compare these prices in different cases as follows. Firstly, when only one retailer offers the return insurance (i.e., $K^* = 01$ or 10), for the retailer with adoption, we have

$$p_L^{10} - p_L^{00} = \frac{2\gamma_H f_L + (2\gamma_H - \gamma_L)(1-\gamma_L)w}{\gamma_L(4\gamma_H - \gamma_L)}, \tag{21}$$

$$p_H^{01} - p_H^{00} = \frac{2\gamma_H f_H + (2\gamma_H - \gamma_L)(1-\gamma_H)w}{\gamma_H(4\gamma_H - \gamma_L)}. \tag{22}$$

Since $0 < \gamma_L < \gamma_H < 1$, $p_L^{10} - p_L^{00} > 0$ and $p_H^{01} - p_H^{00} > 0$ always hold. For the retailer without adoption, we have

$$p_H^{10} - p_H^{00} = -\frac{(1-\gamma_L)w - f_L}{4\gamma_H - \gamma_L}, \text{ and } p_L^{01} - p_L^{00} = -\frac{(1-\gamma_H)w - f_H}{(4\gamma_H - \gamma_L)}. \tag{23}$$

Considering the conditions of the equilibrium cases, we can find that $p_H^{10} - p_H^{00} \le 0$ and $p_L^{01} - p_L^{00} \le 0$ are always satisfied. Specifically, when $K = 10$ emerges as the market equilibrium, we have $\beta \le \frac{(1-\gamma_L)w - f_H}{\gamma_H - \gamma_L}$ and $f_H > (1-\gamma_H)w$. Then, $(1-\gamma_L)w - f_L \ge 0$. Thus, $p_H^{10} - p_H^{00} \le 0$. When $K = 01$ emerges as the market equilibrium, we have $\beta > \frac{(1-\gamma_L)w - f_H}{\gamma_H - \gamma_L}$ and $f_H \le (1-\gamma_H)w$. Then, $p_L^{01} - p_L^{00} \le 0$.

Secondly, when both retailers offer the return insurance (i.e., $K^* = 11$), we have

$$p_L^{11} - p_L^{00} = \frac{2\gamma_H f_L + \gamma_L f_H + (2-\gamma_L)(\gamma_H - \gamma_L)w}{\gamma_L(4\gamma_H - \gamma_L)}, \tag{24}$$

$$p_H^{11} - p_H^{00} = \frac{2\gamma_H f_H + \gamma_H f_L + (1-2\gamma_H)(\gamma_H - \gamma_L)w}{\gamma_H(4\gamma_H - \gamma_L)}. \tag{25}$$

It is easy to find that $p_L^{11} - p_L^{00} > 0$. For $p_H^{11} - p_H^{00}$, we substitute $f_L = f_H + \beta(\gamma_H - \gamma_L)$ into Eq. (25). From $p_H^{11} - p_H^{00} > 0$, we have $\beta > \frac{(2\gamma_H-1)w}{\gamma_H} - \frac{3f_H}{\gamma_H-\gamma_L}$. Then, we find that $\frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L} - [\frac{(2\gamma_H-1)w}{\gamma_H} - \frac{3f_H}{\gamma_H-\gamma_L}] = \frac{2\gamma_H f_H + (2\gamma_H-\gamma_L)(1-\gamma_H)w}{\gamma_H(\gamma_H-\gamma_L)}$, which is always positive. Thus, we can obtain that $\frac{(2\gamma_H-1)w}{\gamma_H} - \frac{3f_H}{\gamma_H-\gamma_L} < \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$.

Proof of Proposition 3. We provide some preliminaries to prove Proposition 3. As discussed in Proof of Proposition 1, retailer L chooses to offer the return insurance when $\Delta_L^{1k_H} \geq \Delta_L^{0k_H}$ (i.e., $(1-\gamma_L)w - f_L \geq 0$), and retailer H chooses to offer the return insurance when $\Delta_H^{k_L 1} \geq \Delta_H^{k_L 0}$ (i.e., $(1-\gamma_H)w - f_H \geq 0$).

Next, given the optimal demand in Eqs. (6), that is $D_L^K = \frac{\gamma_H[(2\gamma_H-\gamma_L)\Delta_L^K - \gamma_L\Delta_H^K]}{\gamma_L(4\gamma_H-\gamma_L)(\gamma_H-\gamma_L)}$ and $D_H^K = \frac{(2\gamma_H-\gamma_L)\Delta_H^K + \gamma_H\Delta_L^K}{(4\gamma_H-\gamma_L)(\gamma_H-\gamma_L)}$, we compare these demand in different cases as follows.

Firstly, we compare the total market size in different cases. From $D_L^K + D_H^K = \frac{2\gamma_H\Delta_L^K + \gamma_L\Delta_H^K}{\gamma_L(4\gamma_H-\gamma_L)}$, we can find that $\frac{\partial(D_L^K + D_H^K)}{\partial\Delta_L^K} = \frac{2\gamma_H}{\gamma_L(4\gamma_H-\gamma_L)} > 0$ and $\frac{\partial(D_L^K + D_H^K)}{\partial\Delta_H^K} = \frac{\gamma_L}{\gamma_L(4\gamma_H-\gamma_L)} > 0$, which mean that as Δ_L^K or Δ_H^K increases, the total market size increases. Then, the adoption of the return insurance enlarges the total market size (i.e., $D_L^{K^*} + D_H^{K^*} \geq D_L^{00} + D_H^{00}$, $K^* = 01$, 10, or 11).

Secondly, we consider the cases when only one retailer offers the return insurance (i.e., $K^* = 01$ or 10). For $K^* = 01$, we have $\Delta_L^{01} = \Delta_L^{00}$ and $\Delta_H^{01} \geq \Delta_H^{00}$. Then, $D_L^{01} \leq D_L^{00}$ and $D_H^{01} \geq D_H^{00}$; For $K^* = 10$, we have $\Delta_L^{10} \geq \Delta_L^{00}$ and $\Delta_H^{10} = \Delta_H^{00}$. Then, $D_L^{10} \geq D_L^{00}$ and $D_H^{10} \leq D_H^{00}$.

Thirdly, when both retailers offer return insurance (i.e., $K = 11$), we have

$$D_L^{11} - D_L^{00} = \frac{\gamma_H\{(2\gamma_H-\gamma_L)[(1-\gamma_L)w - f_L] - \gamma_L[(1-\gamma_H)w - f_H]\}}{\gamma_L(4\gamma_H-\gamma_L)(\gamma_H-\gamma_L)}, \tag{26}$$

$$D_H^{11} - D_H^{00} = \frac{(2\gamma_H-\gamma_L)[(1-\gamma_H)w - f_H] - \gamma_H[(1-\gamma_L)w - f_L]}{(4\gamma_H-\gamma_L)(\gamma_H-\gamma_L)}. \tag{27}$$

From $D_L^{11} - D_L^{00} > 0$, we can obtain that $\frac{(1-\gamma_H)w-f_H}{(1-\gamma_L)w-f_L} < \frac{2\gamma_H-\gamma_L}{\gamma_L}$. Given $f_L = f_H + \beta(\gamma_H - \gamma_L)$, this constraint is updated to $\beta < \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L}$. Moreover, $K = 11$ emerges as the market equilibrium when $\beta \leq \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$ and $f_H \leq (1-\gamma_H)w$. Then, $\frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L} - \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L} = \frac{\gamma_L[(1-\gamma_H)w-f_H]}{(2\gamma_H-\gamma_L)(\gamma_H-\gamma_L)} \geq 0$ always sustains, which means that $\frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L} \leq \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$. Thus, in case $K^* = 11$, (1) if $\beta < \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L}$, $D_L^{11} - D_L^{00} > 0$; (2) if $\frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L} \leq \beta \leq \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$, $D_L^{11} - D_L^{00} \leq 0$.

Similarly, from $D_H^{11} - D_H^{00} > 0$, we can obtain that $\frac{(1-\gamma_H)w-f_H}{(1-\gamma_L)w-f_L} > \frac{\gamma_H}{2\gamma_H-\gamma_L}$, that is $\beta > \frac{(2\gamma_H-1)w+f_H}{\gamma_H}$. Then, we find that $\frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L} - \frac{(2\gamma_H-1)w+f_H}{\gamma_H} = \frac{(2\gamma_H-\gamma_L)[(1-\gamma_H)w-f_H]}{\gamma_H(\gamma_H-\gamma_L)} > 0$. Then, we obtain that (1) if $\beta \leq \frac{f_H+(2\gamma_H-1)w}{\gamma_H}$, $D_H^{11} - D_H^{00} \leq 0$; (2) if $\frac{f_H+(2\gamma_H-1)w}{\gamma_H} < \beta < \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$, $D_H^{11} - D_H^{00} > 0$. Note that $\frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L} - \frac{(2\gamma_H-1)w+f_H}{\gamma_H} = \frac{(4\gamma_H-\gamma_L)w-f_H]}{\gamma_H(2\gamma_H-\gamma_L)} \geq 0$, which means that both $D_L^{11} - D_L^{00} > 0$ and $D_H^{11} - D_H^{00} > 0$ sustain if $\frac{f_H+(2\gamma_H-1)w}{\gamma_H} < \beta < \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L}$.

Proof of Proposition 4. Given the optimal profit in Eq. (7), firstly, we compare the profit in cases of $K = 11$ and $K = 00$.

$$\pi_L^{11} - \pi_L^{00} = \frac{\{(2\gamma_H-\gamma_L)[(1-\gamma_L)w - f_L] - \gamma_L[(1-\gamma_H)w - f_H]\} \cdot B}{\gamma_L(4\gamma_H-\gamma_L)^2(\gamma_H-\gamma_L)}, \tag{28}$$

$$\pi_H^{11} - \pi_H^{00} = \frac{\{[(1-\gamma_H)w - f_H] - \gamma_H[(1-\gamma_L)w - f_L]\} \cdot C}{(4\gamma_H-\gamma_L)^2(\gamma_H-\gamma_L)}, \tag{29}$$

where $B = \gamma_H\{(2\gamma_H-\gamma_L)(\Delta_L^{11} + \Delta_L^{00}) - \gamma_L(\Delta_H^{11} + \Delta_H^{00})\}$ and $C = [(2\gamma_H - \gamma_L)(\Delta_H^{11} + \Delta_H^{00}) + \gamma_H(\Delta_L^{11} + \Delta_L^{00})]$. Given the condition that $\frac{\gamma_H}{2\gamma_H-\gamma_L} < \frac{\Delta_H^K}{\Delta_L^K} < \frac{2\gamma_H-\gamma_L}{\gamma_L}$, we can easily find that $B > 0$ and $C > 0$. Then, $\pi_L^{11} - \pi_L^{00} > 0$ sustains if $(2\gamma_H - \gamma_L)[(1-\gamma_L)w - f_L] - \gamma_L[(1-\gamma_H)w - f_H] > 0$, and $\pi_H^{11} - \pi_H^{00} > 0$ sustains if $[(1-\gamma_H)w - f_H] - \gamma_H[(1-\gamma_L)w - f_L] > 0$. Then, the conditions for Eq. (28) to be positive are the same as those for Eq. (26) to be positive, and the conditions for Eq. (29) to be positive are the same as those for Eq. (27) to be positive. Thus, we can conclude that when $K = 11$ emerges as the market equilibrium: (1) $\pi_L^{11} - \pi_L^{00} > 0$ sustains if $\beta < \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L}$; else if $\frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L} \leq \beta < \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$, $\pi_L^{11} - \pi_L^{00} \leq 0$; (2) $\pi_H^{11} - \pi_H^{00} \leq 0$ sustains if $\beta \leq \frac{f_H+(2\gamma_H-1)w}{\gamma_H}$; else if $\frac{f_H+(2\gamma_H-1)w}{\gamma_H} < \beta < \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$, $\pi_H^{11} - \pi_H^{00} > 0$, where $\frac{f_H+(2\gamma_H-1)w}{\gamma_H} < \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L}$.

Secondly, according to Eqs. (6) and (7), and combine with the conditions under which $K^* = 01$ and 10, we can obtain that $\pi_L^{01} \leq \pi_L^{00}$, $\pi_H^{01} \geq \pi_H^{00}$, $\pi_L^{10} \geq \pi_L^{00}$, and $\pi_H^{10} \leq \pi_H^{00}$, since $D_L^{01} \leq D_L^{00}$, $D_H^{01} \geq D_H^{00}$, $D_L^{10} \geq D_L^{00}$, and $D_H^{10} \leq D_H^{00}$.

Combine with the conditions under which $K = 00$, 01, 10, *or* 11 emerges as the equilibrium, we can conclude that (1) if $\beta \leq \min\{\frac{f_H+(2\gamma_H-1)w}{\gamma_H}, \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}\}$, $\pi_L^{1k_H} \geq \pi_L^{00}$ and $\pi_H^{1k_H} \leq \pi_H^{00}$, $k_H \in \{0,1\}$; (2) if $f_H \leq (1-\gamma_H)w$ and $\frac{f_H+(2\gamma_H-1)w}{\gamma_H} < \beta < \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L}$, $\pi_i^{11} > \pi_i^{00}$, $i \in \{L, H\}$; (3) if $f_H \leq (1-\gamma_H)w$ and $\beta \geq \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L}$, $\pi_L^{k_L 1} \leq \pi_L^{00}$ and $\pi_H^{k_L 1} \geq \pi_H^{00}$, $k_L \in \{0,1\}$; (4) if $f_H > (1-\gamma_H)w$ and $\beta > \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$, $K = 00$ emerges as the equilibrium.

Proof of Proposition 5. For $K = 00$, 01, 10, and 11, from $CS^K = \int_{v_L^K}^{v_{HL}^K} EU_L^K(v)dv + \int_{v_{HL}^K}^1 EU_H^K(v)dv$ and $SW^K = CS^K + \pi_L^K + \pi_H^K$, we obtain the consumer surplus CS^K and the social welfare SW^K as follows.

$$CS^K = \frac{\gamma_H\{(4\gamma_H-3\gamma_L)[\gamma_L(\Delta_H^K)^2 + \gamma_H(\Delta_L^K)^2] - 2(\gamma_L)^2\Delta_H^K\Delta_L^K\}}{2\gamma_L(\gamma_H-\gamma_L)(4\gamma_H-\gamma_L)^2}, \text{ and} \tag{30}$$

$$SW^K = \frac{[12(\gamma_H)^2 - 9\gamma_H\gamma_L + 2(\gamma_L)^2][\gamma_L(\Delta_H^K)^2 + \gamma_H(\Delta_L^K)^2] - 2\gamma_H\gamma_L(8\gamma_H-3\gamma_L)\Delta_H^K\Delta_L^K}{2\gamma_L(\gamma_H-\gamma_L)(4\gamma_H-\gamma_L)^2}. \tag{31}$$

For Eq. (30), we have

$$\frac{\partial CS^K}{\partial\Delta_L^K} = \frac{2\gamma_H\Delta_L^K\{\gamma_H(4\gamma_H-3\gamma_L) - (\gamma_L)^2\frac{\Delta_H^K}{\Delta_L^K}\}}{2\gamma_L(\gamma_H-\gamma_L)(4\gamma_H-\gamma_L)^2}, \tag{32}$$

$$\text{and } \frac{\partial CS^K}{\partial\Delta_H^K} = \frac{\gamma_H\Delta_L^K[(4\gamma_H-3\gamma_L)\frac{\Delta_H^K}{\Delta_L^K} - \gamma_L]}{(\gamma_H-\gamma_L)(4\gamma_H-\gamma_L)^2}. \tag{33}$$

Given the condition that $\frac{\gamma_H}{2\gamma_H-\gamma_L} < \frac{\Delta_H^K}{\Delta_L^K} < \frac{2\gamma_H-\gamma_L}{\gamma_L}$, we find that $\gamma_H(4\gamma_H - 3\gamma_L) - (\gamma_L)^2\frac{\Delta_H^K}{\Delta_L^K} > \gamma_H(4\gamma_H - 3\gamma_L) - (\gamma_L)^2\frac{2\gamma_H-\gamma_L}{\gamma_L} = (4\gamma_H - \gamma_L)(\gamma_H - \gamma_L) > 0$, and $(4\gamma_H - 3\gamma_L)\frac{\Delta_H^K}{\Delta_L^K} - \gamma_L > (4\gamma_H - 3\gamma_L)\frac{\gamma_H}{2\gamma_H-\gamma_L} - \gamma_L = \frac{(4\gamma_H-\gamma_L)(\gamma_H-\gamma_L)}{2\gamma_H-\gamma_L} > 0$. Then, we can conclude that $\frac{\partial CS^K}{\partial\Delta_L^K} > 0$ and $\frac{\partial CS^K}{\partial\Delta_H^K} > 0$ are always satisfied. Recall that retailer L chooses to offer return insurance when $(1-\gamma_L)w - f_L \geq 0$ (i.e., $\Delta_L^{1k_H} \geq \Delta_L^{0k_H}$), and retailer H chooses to offer return insurance when $(1-\gamma_H)w - f_H \geq 0$

(i.e., $\Delta_H^{k_L 1} \geq \Delta_H^{k_L 0}$). Thus, we can obtain that $CS^{11} \geq CS^K \geq CS^{00}$, $K = 01$ or 10.

For Eq. (31), we have

$$SW^{1k_H} - SW^{0k_H} = \frac{\gamma_H[(1-\gamma_L)w - f_L] \cdot E}{2\gamma_L(\gamma_H - \gamma_L)(4\gamma_H - \gamma_L)^2}, \tag{34}$$

$$SW^{k_L 1} - SW^{k_L 0} = \frac{\gamma_L[(1-\gamma_H)w - f_H] \cdot F}{2\gamma_H(\gamma_H - \gamma_L)(4\gamma_H - \gamma_L)^2}, \tag{35}$$

where $E = [12(\gamma_H)^2 - 9\gamma_H\gamma_L + 2(\gamma_L)^2](\Delta_L^{1k_H} + \Delta_L^{0k_H}) - \gamma_L(8\gamma_H - 3\gamma_L)(\Delta_H^{1k_H} + \Delta_H^{0k_H})$ and $F = [12(\gamma_H)^2 - 9\gamma_H\gamma_L + 2(\gamma_L)^2](\Delta_H^{k_L 1} + \Delta_H^{k_L 0}) - \gamma_H(8\gamma_H - 3\gamma_L)(\Delta_L^{k_L 1} + \Delta_L^{k_L 0})$. If $\frac{\Delta_H^K}{\Delta_L^K} \leq \frac{12(\gamma_H)^2 - 9\gamma_H\gamma_L + 2(\gamma_L)^2}{\gamma_L(8\gamma_H - 3\gamma_L)}$, we can obtain that $E \geq 0$ and $SW^{1k_H} - SW^{0k_H} \geq 0$. If $\frac{\Delta_H^K}{\Delta_L^K} \geq \frac{\gamma_H(8\gamma_H - 3\gamma_L)}{12(\gamma_H)^2 - 9\gamma_H\gamma_L + 2(\gamma_L)^2}$, we can obtain that $F \geq 0$ and $SW^{k_L 1} - SW^{k_L 0} \geq 0$. Moreover, $\frac{\gamma_H}{2\gamma_H - \gamma_L} \leq \frac{\Delta_H^K}{\Delta_L^K} \leq \frac{2\gamma_H - \gamma_L}{\gamma_L}$ should be satisfied to ensure that both retailers have positive sales. Then, we find that $\frac{\gamma_H(8\gamma_H - 3\gamma_L)}{12(\gamma_H)^2 - 9\gamma_H\gamma_L + 2(\gamma_L)^2} - \frac{\gamma_H}{2\gamma_H - \gamma_L} = \frac{(4\gamma_H - \gamma_L)(\gamma_H - \gamma_L)}{[12(\gamma_H)^2 - 9\gamma_H\gamma_L + 2(\gamma_L)^2](2\gamma_H - \gamma_L)} > 0$, $\frac{2\gamma_H - \gamma_L}{\gamma_L} - \frac{12(\gamma_H)^2 - 9\gamma_H\gamma_L + 2(\gamma_L)^2}{\gamma_L(8\gamma_H - 3\gamma_L)} = \frac{(4\gamma_H - \gamma_L)(\gamma_H - \gamma_L)}{\gamma_L(8\gamma_H - 3\gamma_L)} > 0$. This means that $[\frac{\gamma_H(8\gamma_H - 3\gamma_L)}{12(\gamma_H)^2 - 9\gamma_H\gamma_L + 2(\gamma_L)^2}, \frac{12(\gamma_H)^2 - 9\gamma_H\gamma_L + 2(\gamma_L)^2}{\gamma_L(8\gamma_H - 3\gamma_L)}] \subset (\frac{\gamma_H}{2\gamma_H - \gamma_L}, \frac{2\gamma_H - \gamma_L}{\gamma_L})$.

Appendix B. Proof of results in extensions

B.1. Online retailers endogenously determine the quality levels.

Lemma 4. *When the quality levels of two competing retailers are identical, they both adopt or abandon the return insurance and their payoffs are zero. Moreover, the two competing retailers are better off by differentiating their quality levels.*

Lemma 4 shows that two competing retailers always differentiate their quality levels. When the quality levels of two competing retailers are identical, their satisfaction rates are also the same and consumers make purchasing decisions by comparing the retail prices. Then, the retailer who charges a lower price than the price of his competitor can extract all the market shares and his competitor has no sales. As long as the price is higher than the marginal cost, each retailer has an incentive to deviate to a lower price than his competitor to grab more market shares. Thus, both retailers charge identical prices which are equal to the marginal cost, and earn zero profit. Moreover, owing to the same satisfaction rate, both retailers adopt return insurance if this adoption is efficient within the satisfaction rate; otherwise, both retailers abandon the return insurance. In contrast, when retailers offer different quality levels, both retailers can gain positive profit. This can be inferred in Eq. (7) in the basic model of two competing retailers with different satisfaction rates. Thus, the two competing retailers are better off by differentiating their quality levels (satisfaction rates).

Lemma 5. *When two competing retailers differentiate their quality levels, if* $(1-\gamma_i)w - f(\gamma_i) \geq 0$ *sustains for any* $\gamma_i \in (0,1)$*, both retailers offer return insurance.*

Lemma 5 demonstrates that for any quality level, if the return insurance is efficient (i.e., $(1-\gamma_i)w - f(\gamma_i) \geq 0$), both retailers offer return insurance. This coincides with Proposition 1, the insurance reduces the consumers' hassle cost and more consumers are willing to make purchases. Then, as long as the additional profit brought by the insurance can counterbalance the cost of premiums, both retailers offer return insurance.

Proof of Lemma 4. The case in which two competing retailers set the same quality levels is the same as that studied in Huang et al. (2018). In this case, if the prices of retailer L and retailer H are identical, the consumers' expected utilities of purchasing a product from retailer L and retailer H are equal owing to the same satisfaction rates (i.e., $\gamma_H = \gamma_L$). That is if $p_L^K = p_H^K$, $EU_L^K = EU_H^K$. Then, following the analysis in Appendix A, as long as the return insurance is efficient for retailers with a satisfaction rate γ_i (i.e., $(1-\gamma_i)w - f(\gamma_i) \geq 0$), both retailers adopt the return insurance; otherwise, both retailers abandon the return insurance.

Moreover, regarding the retailers' price decisions, we can find that retailer i who charges a lower price than the price of his competitor (i.e., $p_i^K < p_{-i}^K$) can extract all the market shares and his competitor has no sales (i.e., $D_i = 1 - \frac{\gamma_i p_i + (1-\gamma_i)(r - k_i w)}{\gamma_i}$ and $D_{-i} = 0$). Therefore, as long as the price is higher than the marginal cost, each retailer has incentives to deviate to a lower price than his competitor to grab more market shares. Hence, the case that both retailers charge the same prices at the marginal costs emerges as the equilibrium. In this case, both retailers earn zero profit.

Furthermore, from Eq. (7) in the basic model, we can easily find that with the different quality levels (satisfaction rates), both retailers can gain positive profit. Thus, the scenario where retailers differentiate the quality levels dominates the scenario where retailers offer products with the same quality levels. In other words, two competing retailers always differentiate their quality levels.

Proof of Lemma 5. As shown in Proof of Proposition 1 in Appendix A, we have $\pi_L^{1k_H} - \pi_L^{0k_H} \geq 0$ if $(1-\gamma_L)w - f_L \geq 0$, and $\pi_H^{k_L 1} - \pi_H^{k_L 0} \geq 0$ if $(1-\gamma_H)w - f_H \geq 0$. Thus, for any γ_i, if $(1-\gamma_i)w - f(\gamma_i) \geq 0$ is always satisfied, the return insurance is efficient for retailers with satisfaction rate γ_i and both retailers offer return insurance.

Table B.1
Two retailers' decisions on satisfaction rates.

(γ_H, γ_L)	$w = 0$	$w = 0.015$	$w = 0.03$	$w = 0.045$
(a) *Convex* : $f(\gamma_i) = \xi(1 - \sqrt{\gamma_i})$				
$\xi/w = 0.2$	(0.8684, 0.3926)	(0.8372, 0.3543)	(0.8011, 0.3077)	(0.7503, 0.2327)
$\xi/w = 0.4$	(0.8684, 0.3926)	(0.8419, 0.3610)	(0.8127, 0.3249)	(0.7784, 0.2797)
$\xi/w = 0.6$	(0.8684, 0.3926)	(0.8465, 0.3675)	(0.8233, 0.3402)	(0.7980, 0.3095)
$\xi/w = 0.8$	(0.8684, 0.3926)	(0.8510, 0.3737)	(0.8330, 0.3540)	(0.8145, 0.3333)
$\xi/w = 1$	(0.8684, 0.3926)	(0.8533, 0.3797)	(0.8423, 0.3668)	(0.8291, 0.3538)
(b) *Concave* : $f(\gamma_i) = \xi(1 - \gamma_i^2)$				
$\xi/w = 0.1$	(0.8684, 0.3926)	(0.8376, 0.3529)	(0.8008, 0.3026)	(0.7405, 0.1987)
$\xi/w = 0.2$	(0.8684, 0.3926)	(0.8429, 0.3584)	(0.8131, 0.3166)	(0.7719, 0.2537)
$\xi/w = 0.3$	(0.8684, 0.3926)	(0.8483, 0.3639)	(0.8249, 0.3297)	(0.7954, 0.2844)
$\xi/w = 0.4$	(0.8684, 0.3926)	(0.8535, 0.3692)	(0.8364, 0.3421)	(0.8157, 0.3086)
$\xi/w = 0.5$	(0.8684, 0.3926)	(0.8588, 0.3745)	(0.8477, 0.3539)	(0.8346, 0.3297)
(c) *Linear* : $f(\gamma_i) = \xi(1 - \gamma_i)$				
$\xi/w = 0.2$	(0.8684, 0.3926)	(0.8399, 0.3571)	(0.8072, 0.3142)	(0.7636, 0.2506)
$\xi/w = 0.4$	(0.8684, 0.3926)	(0.8474, 0.3665)	(0.8243, 0.3370)	(0.7979, 0.3015)
$\xi/w = 0.6$	(0.8684, 0.3926)	(0.8546, 0.3755)	(0.8399, 0.3571)	(0.8243, 0.3370)
$\xi/w = 0.8$	(0.8684, 0.3926)	(0.8616, 0.3842)	(0.8546, 0.3755)	(0.8474, 0.3665)
$\xi/w = 1$	(0.8684, 0.3926)	(0.8684, 0.3926)	(0.8684, 0.3926)	(0.8684, 0.3926)

Results of numerical experiments.

(1) **Two retailers' decisions on satisfaction rates when $(1-\gamma_i)\mathbf{w} - \mathbf{f}(\gamma_i) \geq 0$ is satisfied for any** γ_i. In this case, both retailers offer return insurance, as demonstrated in Lemma 5. Note that (a) for $f(\gamma_i) = \xi(1 - \sqrt{\gamma_i})$ ($f''(\gamma_i) > 0$), $(1-\gamma_i)w - f(\gamma_i) \geq 0$ always holds when $\xi/w \leq 1$; (b) for $f(\gamma_i) = \xi(1 - \gamma_i^2)$ ($f''(\gamma_i) < 0$), $(1-\gamma_i)w - f(\gamma_i) \geq 0$ always holds when $\xi/w \leq 0.5$; (c) for $f(\gamma_i) = \xi(1-\gamma_i)$ ($f''(\gamma_i) = 0$), $(1-\gamma_i)w - f(\gamma_i) \geq 0$ always holds when $\xi/w \leq 1$. Moreover, the case when no retailer offers return insurance (i.e., $K^* = 00$) is the same as the case of $w = 0$ in the following tables. Then, given $\alpha = 0.45$, and $S_i - T_i - r = 0.02$, the results of two retailers' decisions on satisfaction rates are shown in the following Table B.1.

(2) **Two retailers' decisions on satisfaction rates and the equilibrium of two retailers' return insurance strategies when $(1-\gamma_i)\mathbf{w} - \mathbf{f}(\gamma_i) \geq 0$ is satisfied in certain conditions**. Note that (a) for

Table B.2
Two retailers' decisions on satisfaction rates under the convex function: $f(\gamma_i) = \xi(1-\sqrt{\gamma_i})$

	$\xi/w = 1.6$			$\xi/w = 1.65$	
(γ_L, γ_H)	$k_H = 0$	$k_H = 1$	(γ_L, γ_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.3926, 0.8684)	(0.3905, 0.8644)	$k_L = 0$	(0.3926,0.8684)	(0.3909, 0.8651)
$k_L = 1$	(0.3987, 0.8718)	(0.3967, 0.8678)	$k_L = 1$	(0.3997,0.8721)	(0.3981, 0.8689)
(π_L, π_H)	$k_H = 0$	$k_H = 1$	(π_L, π_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.028702, 0.039697)	(0.028515, 0.039979)	$k_L = 0$	(0.028702, 0.039697)	(0.028547, 0.039928)
$k_L = 1$	(0.028887, 0.039106)	(0.028697, 0.039376)	$k_L = 1$	(0.028811, 0.039059)	(0.028657, 0.039278)
	Equilibrium: $K^* = 11$			Equilibrium: $K^* = 11$	
	$\xi/w = 1.7$			$\xi/w = 1.75$	
(γ_L, γ_H)	$k_H = 0$	$k_H = 1$	(γ_L, γ_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.3926, 0.8684)	(0.3912, 0.8658)	$k_L = 0$	(0.3926,0.8684)	(0.3916, 0.8666)
$k_L = 1$	(0.4007, 0.8724)	(0.3994, 0.8698)	$k_L = 1$	(0.4017,0.8727)	(0.4008, 0.8709)
(π_L, π_H)	$k_H = 0$	$k_H = 1$	(π_L, π_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.028702, 0.039697)	(0.028579, 0.039887)	$k_L = 0$	(0.028702, 0.039697)	(0.028616, 0.039837)
$k_L = 1$	(0.028735, 0.039011)	(0.028610, 0.039190)	$k_L = 1$	(0.028660, 0.038963)	(0.028571, 0.039092)
	Equilibrium: $K^* = 11$			Equilibrium: $K^* = 01$	
	$\xi/w = 1.8$			$\xi/w = 1.85$	
(γ_L, γ_H)	$k_H = 0$	$k_H = 1$	(γ_L, γ_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.3926, 0.8684)	(0.3920, 0.8673)	$k_L = 0$	(0.3926,0.8684)	(0.3923, 0.8680)
$k_L = 1$	(0.4027, 0.8730)	(0.4021, 0.8718)	$k_L = 1$	(0.4036,0.8732)	(0.4034, 0.8728)
(π_L, π_H)	$k_H = 0$	$k_H = 1$	(π_L, π_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.028702, 0.039697)	(0.028647, 0.039787)	$k_L = 0$	(0.028702, 0.039697)	(0.028679, 0.039746)
$k_L = 1$	(0.028585, 0.038915)	(0.028525, 0.039003)	$k_L = 1$	(0.028507, 0.038876)	(0.028483, 0.038915)
	Equilibrium: $K^* = 01$			Equilibrium: $K^* = 01$	
	$\xi/w = 1.9$			$\xi/w = 1.95$	
(γ_L, γ_H)	$k_H = 0$	$k_H = 1$	(γ_L, γ_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.3926, 0.8684)	(0.3927, 0.8688)	$k_L = 0$	(0.3926,0.8684)	(0.3930, 0.8695)
$k_L = 1$	(0.4046, 0.8735)	(0.4047, 0.8738)	$k_L = 1$	(0.4056,0.8737)	(0.4060, 0.8747)
(π_L, π_H)	$k_H = 0$	$k_H = 1$	(π_L, π_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.028702, 0.039697)	(0.028715, 0.039696)	$k_L = 0$	(0.028702, 0.039697)	(0.028747, 0.039656)
$k_L = 1$	(0.028433, 0.0388275)	(0.028442, 0.038827)	$k_L = 1$	(0.028355, 0.038779)	(0.028396, 0.038738)
	Equilibrium: $K^* = 00$			Equilibrium: $K^* = 00$	

$f(\gamma_i) = \xi(1-\sqrt{\gamma_i})$ ($f''(\gamma_i) > 0$), when $1 < \xi/w \le 2$, $(1-\gamma_i)w - f(\gamma_i) \ge 0$ holds if $\gamma_i \in [(\xi/w-1)^2$; (b) for $f(\gamma_i) = \xi(1-\gamma_i^2)$ ($f''(\gamma_i) < 0$), when $0.5 < \xi/w \le 1$, $(1-\gamma_i)w - f(\gamma_i) \ge 0$ holds if $\gamma_i \in (0, \xi/w - 1]$; (c) for $f(\gamma_i) = \xi(1-\gamma_i)$ ($f''(\gamma_i) = 0$), $(1-\gamma_i)w - f(\gamma_i) \ge 0$ always holds or never holds. Then, given $\alpha = 0.45$, $S_i - T_i - r = 0.02$, and $w = 0.015$, the results are shown in the following Tables B.2 and B.3. Note that the upper part of each sub-table shows the retailers' optimal satisfaction rates in four cases (i.e., $K \in \{00, 01, 10, 11\}$), and the lower part shows the two retailers' payment matrix. As a result, at the bottom of each sub-table we provide the optimal strategy equilibrium under each numerical experiment.

(3) **The decisions on the satisfaction rates and prices, demand, profit, equilibrium strategies, the consumer surplus, and the social welfare.** Given $\alpha = 0.45$, $S_i - T_i - r = 0.02$, and $w = 0.015$, the results are shown in the following Tables B.4–B.6.

B.2. Consumers' valuation is influenced by the quality.

Proof of Lemma 2. Please refer to Proof of Lemma 1, following the same procedure, we drive the solutions of the retailers' optimal pricing strategies in four cases (i.e., $K = 00, 01, 10$, and 11) and show these results in Lemma 2.

Moreover, from Eq. (12) in Lemma 2, we can find that both retailers have positive sales (co-exist) if and only if $\hat{D}_L^K > 0$ and $\hat{D}_H^K > 0$. From $\hat{D}_L^K > 0$, we obtain that $\frac{\hat{\Delta}_H^K}{\hat{\Delta}_L^K} < \frac{2\gamma_H q_H - \gamma_L q_L}{\gamma_L q_L}$. From $\hat{D}_H^K > 0$, $\frac{\hat{\Delta}_H^K}{\hat{\Delta}_L^K} > \frac{\gamma_H q_H}{2\gamma_H q_H - \gamma_L q_L}$. To sum up, we have

(1) if $\frac{\hat{\Delta}_H^K}{\hat{\Delta}_L^K} \le \frac{\gamma_H q_H}{2\gamma_H q_H - \gamma_L q_L}$, retailer H has no sales;

(2) if $\frac{\gamma_H q_H}{2\gamma_H q_H - \gamma_L q_L} < \frac{\hat{\Delta}_H^K}{\hat{\Delta}_L^K} < \frac{2\gamma_H q_H - \gamma_L q_L}{\gamma_L q_L}$, both retailers H and L have positive sales;

(3) if $\frac{\hat{\Delta}_H^K}{\hat{\Delta}_L^K} \ge \frac{2\gamma_H q_H - \gamma_L q_L}{\gamma_L q_L}$, retailer L has no sales.

Proof of Lemma 3. From Table 6, we can easily find that $\hat{\Delta}_L^{k_L 0} = \hat{\Delta}_L^{k_L 1}$, $\hat{\Delta}_H^{0k_H} = \hat{\Delta}_H^{1k_H}$, $\hat{\Delta}_L^{1k_H} - \hat{\Delta}_L^{0k_H} = (1-\gamma_L)w - f_L$, and $\hat{\Delta}_H^{k_L 1} - \hat{\Delta}_H^{k_L 0} = (1-\gamma_H)w - f_H$, where $k_L \in \{0,1\}$ and $k_H \in \{0,1\}$. Moreover, we have

$$\hat{\pi}_L^{1k_H} - \hat{\pi}_L^{0k_H} = \frac{[(1-\gamma_L)w - f_L] \cdot \gamma_H q_H (2\gamma_H q_H - \gamma_L q_L) \cdot \widehat{A_1}}{\gamma_L q_L (4\gamma_H q_H - \gamma_L q_L)^2 (\gamma_H q_H - \gamma_L q_L)}, \text{ and} \quad (36)$$

$$\hat{\pi}_H^{k_L 1} - \hat{\pi}_H^{k_L 0} = \frac{[(1-\gamma_H)w - f_H] \cdot (2\gamma_H q_H - \gamma_L q_L) \cdot \widehat{A_2}}{(4\gamma_H q_H - \gamma_L q_L)^2 (\gamma_H q_H - \gamma_L q_L)}, \quad (37)$$

where $\widehat{A_1} = (2\gamma_H q_H - \gamma_L q_L)(\hat{\Delta}_L^{1k_H} + \hat{\Delta}_L^{0k_H}) - \gamma_L q_L (\hat{\Delta}_H^{1k_H} + \hat{\Delta}_H^{0k_H})$, and $\widehat{A_2} = (2\gamma_H q_H - \gamma_L q_L)(\hat{\Delta}_H^{k_L 1} + \hat{\Delta}_H^{k_L 0}) - \gamma_H q_H (\hat{\Delta}_L^{k_L 1} + \hat{\Delta}_L^{k_L 0})$. Given the condition that $\frac{\gamma_H q_H}{2\gamma_H q_H - \gamma_L q_L} < \frac{\hat{\Delta}_H^K}{\hat{\Delta}_L^K} < \frac{2\gamma_H q_H - \gamma_L q_L}{\gamma_L q_L}$, we can easily find that $\widehat{A_1} > 0$ and $\widehat{A_2} > 0$. Thus, $\hat{\pi}_L^{1k_H} - \hat{\pi}_L^{0k_H} \ge 0$ sustains if $(1-\gamma_L)w - f_L \ge 0$, and $\hat{\pi}_H^{k_L 1} - \hat{\pi}_H^{k_L 0} \ge 0$ sustains if $(1-\gamma_H)w - f_H \ge 0$.

Proof of Proposition 6. Given the optimal profit in Eq. (13), firstly, we compare the profit in cases of $K = 11$ and $K = 00$.

$$\hat{\pi}_L^{11} - \hat{\pi}_L^{00} = \frac{\{(2\gamma_H q_H - \gamma_L q_L)[(1-\gamma_L)w - f_L] - \gamma_L q_L [(1-\gamma_H)w - f_H]\} \cdot \hat{B}}{\gamma_L q_L (4\gamma_H q_H - \gamma_L q_L)^2 (\gamma_H q_H - \gamma_L q_L)}, \quad (38)$$

$$\hat{\pi}_H^{11} - \hat{\pi}_H^{00} = \frac{\{(2\gamma_H q_H - \gamma_L q_L)[(1-\gamma_H)w - f_H] - \gamma_H q_H [(1-\gamma_L)w - f_L]\} \cdot \hat{C}}{(4\gamma_H q_H - \gamma_L q_L)^2 (\gamma_H q_H - \gamma_L q_L)}, \quad (39)$$

where $\hat{B} = \gamma_H q_H \{(2\gamma_H q_H - \gamma_L q_L)(\hat{\Delta}_L^{11} + \hat{\Delta}_L^{00}) - \gamma_L q_L (\hat{\Delta}_H^{11} + \hat{\Delta}_H^{00})\}$ and $\hat{C} = [(2\gamma_H q_H - \gamma_L q_L)(\hat{\Delta}_H^{11} + \hat{\Delta}_H^{00}) + \gamma_H q_H (\hat{\Delta}_L^{11} + \hat{\Delta}_L^{00})]$. Given the condition that $\frac{\gamma_H q_H}{2\gamma_H q_H - \gamma_L q_L} < \frac{\hat{\Delta}_H^K}{\hat{\Delta}_L^K} < \frac{2\gamma_H q_H - \gamma_L q_L}{\gamma_L q_L}$, we can easily find that $\hat{B} > 0$

Table B.3
Two retailers' decisions on satisfaction rates under the concave function: $f(\gamma_i) = \xi(1-\gamma_i^2)$.

	$\xi/w = 0.55$			$\xi/w = 0.6$	
(γ_L, γ_H)	$k_H = 0$	$k_H = 1$	(γ_L, γ_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.3926, 0.8684)	(0.3922, 0.8674)	$k_L = 0$	(0.3926,0.8684)	(0.3935, 0.8700)
$k_L = 1$	(0.4007, 0.8724)	(0.3771, 0.8614)	$k_L = 1$	(0.3789,0.8626)	(0.3797, 0.8640)
(π_L, π_H)	$k_H = 0$	$k_H = 1$	(π_L, π_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.028702, 0.039697)	(0.028669, 0.039721)	$k_L = 0$	(0.028702, 0.039697)	(0.028787, 0.039554)
$k_L = 1$	(0.029511, 0.038616)	(0.029148, 0.040759)	$k_L = 1$	(0.028980, 0.040711)	(0.029054, 0.040575)
	Equilibrium: $K^* = 11$			Equilibrium: $K^* = 10$	
	$\xi/w = 0.65$			$\xi/w = 0.7$	
(γ_L, γ_H)	$k_H = 0$	$k_H = 1$	(γ_L, γ_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.3926, 0.8684)	(0.3946, 0.8723)	$k_L = 0$	(0.3926,0.8684)	(0.3959, 0.8749)
$k_L = 1$	(0.3802, 0.8627)	(0.3823, 0.8666)	$k_L = 1$	(0.3776,0.8626)	(0.3849, 0.8692)
(π_L, π_H)	$k_H = 0$	$k_H = 1$	(π_L, π_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.028702, 0.039697)	(0.028893, 0.039407)	$k_L = 0$	(0.028702, 0.039697)	(0.029010, 0.039244)
$k_L = 1$	(0.028776, 0.040695)	(0.028962, 0.040393)	$k_L = 1$	(0.028564, 0.041040)	(0.028871, 0.040211)
	Equilibrium: $K^* = 10$			Equilibrium: $K^* = 00$	
	$\xi/w = 0.75$			$\xi/w = 1.85$	
(γ_L, γ_H)	$k_H = 0$	$k_H = 1$	(γ_L, γ_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.3926, 0.8684)	(0.3971, 0.8774)	$k_L = 0$	(0.3926,0.8684)	(0.3983, 0.8798)
$k_L = 1$	(0.3828, 0.8629)	(0.3874, 0.8718)	$k_L = 1$	(0.3841,0.8630)	(0.3900, 0.8744)
(π_L, π_H)	$k_H = 0$	$k_H = 1$	(π_L, π_H)	$k_H = 0$	$k_H = 1$
$k_L = 0$	(0.028702, 0.039697)	(0.029122, 0.039090)	$k_L = 0$	(0.028702, 0.039697)	(0.029231, 0.038938)
$k_L = 1$	(0.028372, 0.040665)	(0.028780, 0.040040)	$k_L = 1$	(0.028172, 0.040649)	(0.028691, 0.039862)
	Equilibrium: $K^* = 00$			Equilibrium: $K^* = 00$	

Table B.4
Equilibrium results under the convex function: $f(\gamma_i) = \xi(1-\sqrt{\gamma_i})$.

ξ/w	Equilibrium	γ_L	γ_H	p_L	p_H	D_L	D_H	π_L	π_H	CS	SW
0.2	$K^* = 11$	0.3543	0.8372	0.219937	0.530574	0.383313	0.296514	0.030027	0.042457	0.103101	0.175585
0.4	$K^* = 11$	0.3610	0.8419	0.227001	0.531251	0.380289	0.295356	0.029821	0.041951	0.103373	0.175146
0.6	$K^* = 11$	0.3675	0.8465	0.233690	0.531945	0.377424	0.294226	0.029623	0.041467	0.103625	0.174715
0.8	$K^* = 11$	0.3737	0.8510	0.239978	0.532692	0.374716	0.293129	0.029430	0.041012	0.103844	0.174286
1	$K^* = 11$	0.3797	0.8533	0.245457	0.532312	0.371984	0.292708	0.029161	0.040577	0.104267	0.173905
1.3	$K^* = 11$	0.3884	0.8617	0.254366	0.534534	0.368474	0.290554	0.028965	0.039957	0.104323	0.173245
1.5	$K^* = 11$	0.3940	0.8658	0.259666	0.535267	0.366160	0.289579	0.028786	0.039563	0.104491	0.172840
1.55	$K^* = 11$	0.3954	0.8669	0.260987	0.535490	0.365602	0.289312	0.028745	0.039465	0.104528	0.172739
1.6	$K^* = 11$	0.3967	0.8678	0.262206	0.535640	0.365041	0.289109	0.028697	0.039376	0.104565	0.172638
1.65	$K^* = 11$	0.3981	0.8689	0.263510	0.535862	0.364493	0.288841	0.028657	0.039278	0.104603	0.172539
1.7	$K^* = 11$	0.3994	0.8698	0.264714	0.536012	0.363942	0.288638	0.028610	0.039190	0.104639	0.172439
1.75	$K^* = 01$	0.3916	0.8666	0.232653	0.536943	0.365123	0.289598	0.028616	0.039837	0.103850	0.172302
1.8	$K^* = 01$	0.3920	0.8673	0.236931	0.537210	0.365173	0.289325	0.028647	0.039787	0.103853	0.172287
1.85	$K^* = 01$	0.3923	0.8680	0.237280	0.537512	0.365230	0.289055	0.028679	0.039746	0.103843	0.172268
1.9	$K^* = 00$	0.3926	0.8684	0.237570	0.535402	0.365285	0.288847	0.028702	0.039697	0.103843	0.172243
1.95	$K^* = 00$	0.3926	0.8684	0.237570	0.535402	0.365285	0.288847	0.028702	0.039697	0.103843	0.172243

Table B.5
Equilibrium results under the concave function: $f(\gamma_i) = \xi(1-\gamma_i^2)$.

ξ/w	Equilibrium	γ_L	γ_H	p_L	p_H	D_L	D_H	π_L	π_H	CS	SW
0.1	$K^* = 11$	0.3529	0.8376	0.219390	0.531472	0.383450	0.296308	0.030027	0.042556	0.102811	0.175393
0.2	$K^* = 11$	0.3584	0.8429	0.226160	0.533070	0.380481	0.294899	0.029823	0.042135	0.102807	0.174765
0.3	$K^* = 11$	0.3639	0.8483	0.232783	0.534708	0.377603	0.293474	0.029628	0.041720	0.102800	0.174148
0.4	$K^* = 11$	0.3692	0.8535	0.239086	0.536097	0.374815	0.292128	0.029431	0.041330	0.102777	0.173538
0.5	$K^* = 11$	0.3745	0.8588	0.245264	0.537925	0.372108	0.290766	0.029242	0.040945	0.102750	0.172938
0.55	$K^* = 11$	0.3771	0.8614	0.248261	0.538725	0.370783	0.290106	0.029148	0.040759	0.102734	0.172641
0.6	$K^* = 10$	0.3789	0.8626	0.250412	0.536514	0.369319	0.290112	0.028980	0.040711	0.102737	0.172428
0.65	$K^* = 10$	0.3802	0.8627	0.252055	0.536319	0.367869	0.290418	0.028776	0.040695	0.102725	0.172197
0.7	$K^* = 00$	0.3926	0.8684	0.23757	0.535402	0.365285	0.288847	0.028702	0.039697	0.103843	0.172243
0.75	$K^* = 00$	0.3926	0.8684	0.23757	0.535402	0.365285	0.288847	0.028702	0.039697	0.103843	0.172243
0.8	$K^* = 00$	0.3926	0.8684	0.23757	0.535402	0.365285	0.288847	0.028702	0.039697	0.103843	0.172243

Table B.6
Equilibrium results under the linear function: $f(\gamma_i) = \xi(1-\gamma_i)$.

ξ/w	Equilibrium	γ_L	γ_H	p_L	p_H	D_L	D_H	π_L	π_H	CS	SW
0.2	$K^* = 11$	0.3571	0.8399	0.223473	0.531416	0.381691	0.295817	0.029906	0.042249	0.103082	0.175236
0.4	$K^* = 11$	0.3665	0.8474	0.233791	0.533015	0.377199	0.293942	0.029593	0.041551	0.103317	0.174460
0.6	$K^* = 11$	0.3755	0.8546	0.243366	0.534584	0.372991	0.292171	0.029287	0.040898	0.103517	0.173702
0.8	$K^* = 11$	0.3842	0.8616	0.252344	0.536137	0.369029	0.290473	0.028990	0.040280	0.103692	0.172963
1	$K^* = 00$	0.3926	0.8684	0.260777	0.537675	0.365285	0.288847	0.028702	0.039697	0.103843	0.172243

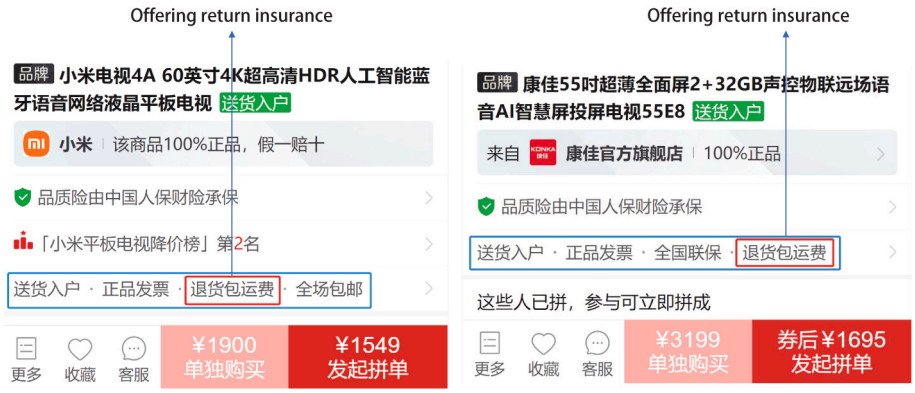


Fig. 5. The return insurance strategies of Xiaomi and KONKA on the PDD app.

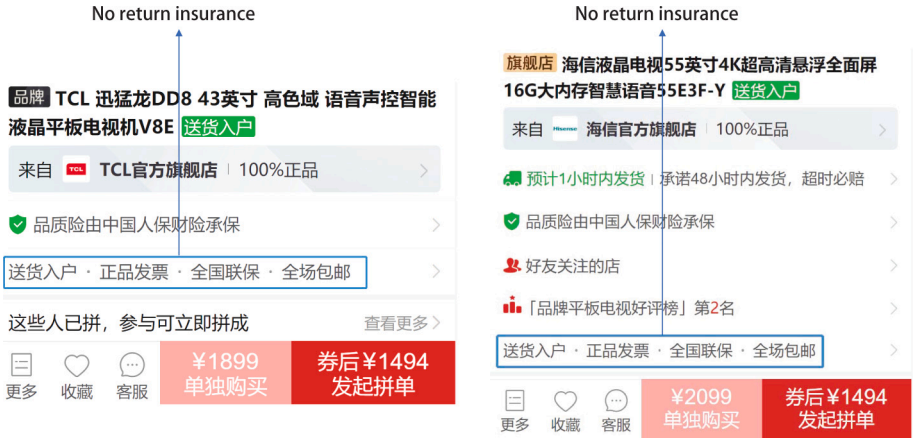


Fig. 6. The return insurance strategies of TCL and Hisense on the PDD app.

and $\hat{C} > 0$. Then, $\hat{\pi}_L^{11} - \hat{\pi}_L^{00} > 0$ sustains if $\frac{(1-\gamma_H)w-f_H}{(1-\gamma_L)w-f_L} < \frac{2\gamma_H q_H - \gamma_L q_L}{\gamma_L q_L}$, and $\hat{\pi}_H^{11} - \hat{\pi}_H^{00} > 0$ sustains if $\frac{(1-\gamma_H)w-f_H}{(1-\gamma_L)w-f_L} > \frac{\gamma_H q_H}{2\gamma_H q_H - \gamma_L q_L}$. Thus, we can conclude that when $K = 11$ emerges as the market equilibrium: (1) $\hat{\pi}_L^{11} - \hat{\pi}_L^{00} > 0$ sustains if $\beta < \overline{\beta}$, where $\overline{\beta} = w + \frac{(1-\gamma_H)w-f_H}{\gamma_H-\gamma_L}[1 - \frac{\gamma_L}{2\gamma_H \cdot (\frac{q_H}{q_L}) - \gamma_L}]$; else if $\overline{\beta} \le \beta < \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$, $\hat{\pi}_L^{11} - \hat{\pi}_L^{00} \le 0$; (2) $\hat{\pi}_H^{11} - \hat{\pi}_H^{00} \le 0$ sustains if $\beta \le \underline{\beta}$, where $\underline{\beta} = w - \frac{(1-\gamma_H)w-f_H}{\gamma_H-\gamma_L}[1 - \frac{\gamma_L}{\gamma_H \cdot (\frac{q_H}{q_L})}]$; else if $\underline{\beta} < \beta < \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$, $\hat{\pi}_H^{11} - \hat{\pi}_H^{00} > 0$, where $\underline{\beta} < \overline{\beta}$. Note that when $\frac{q_H}{q_L} = 1$, $\underline{\beta} = \frac{f_H + (2\gamma_H - 1)w}{\gamma_H}$ and $\overline{\beta} = \frac{(2-\gamma_L)w-2f_H}{2\gamma_H-\gamma_L}$.

Secondly, according to Eqs. (12) and (13), and combine with the conditions under which $K = 01$ and 10, we can obtain that $\hat{\pi}_L^{01} \le \hat{\pi}_L^{00}$, $\hat{\pi}_H^{01} \ge \hat{\pi}_H^{00}$, $\hat{\pi}_L^{10} \ge \hat{\pi}_L^{00}$, and $\hat{\pi}_H^{10} \le \hat{\pi}_H^{00}$.

To sum up, combine with the conditions under which $K = 00$, 01, 10, *or* 11 emerges as the equilibrium, we can conclude that (1) if $\beta \le \min\{\underline{\beta}, \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}\}$, $\hat{\pi}_L^{1k_H} \ge \hat{\pi}_L^{00}$ and $\hat{\pi}_H^{1k_H} \le \hat{\pi}_H^{00}$, $k_H \in \{0,1\}$; (2) if $f_H \le (1-\gamma_H)w$ and $\underline{\beta} < \beta < \overline{\beta}$, $\pi_i^{11} > \pi_i^{00}$, $i \in \{L,H\}$; (3) if $f_H \le (1-\gamma_H)w$ and $\beta \ge \overline{\beta}$, $\hat{\pi}_L^{k_L 1} \le \hat{\pi}_L^{00}$ and $\hat{\pi}_H^{k_L 1} \ge \hat{\pi}_H^{00}$, $k_L \in \{0,1\}$; (4) if $f_H > (1-\gamma_H)w$ and $\beta > \frac{(1-\gamma_L)w-f_H}{\gamma_H-\gamma_L}$, $K = 00$ emerges as the equilibrium.

Appendix C. Figures of retailers' return insurance strategies.

In Fig. 5, Xiaomi and KONKA purchase return insurance for consumers.

In Fig. 6, TCL and Hisense do not offer return insurance for consumers.

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